An explicit exponent for the exceptional set of the Erdős–Straus equation

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The Erdős–Straus conjecture says that four divided by any integer greater than one is a sum of three unit fractions. In 1970 Vaughan showed that the integers up to N for which this fails are few: their number is at most N times the exponential of minus a positive constant times the two-thirds power of $\log N$*, and his argument gives no value of the constant. We prove this bound with an explicit constant, given by an integral whose value is* 0.6792 *and which is at least* 0.6657 *in closed form. The residue classes we sieve with include classes to moduli beyond the square root of each prime, which a recent theorem of Maynard lets us count.*

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CONTENTS

I. Introduction 1
II. Type II triples and their classes 3
III. Distinct classes in the corners 3
IV. The centre 5
V. Counting the classes of the corners 6
VI. The strips of the centre 7
VII. The large sieve and the tilt 10
VIII. Proofs of the theorems 10
IX. The value of the constant 11
X. Remarks 11
XI. Formal and numerical verification 13
A. Detailed derivations 13
Data availability 18
Funding 18
Competing interests 18
References 18

I. INTRODUCTION

For an integer $n \geq 2$ the Erdős–Straus conjecture asserts that

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z} \tag{1}$$

has a solution in positive integers x, y, z [11, §D11]. A solution for n gives one for every multiple of n, so it is enough to treat primes, and polynomial identities settle every prime outside the six residue classes $1, 121, 169, 289, 361, 529$ modulo 840 [19]. These six classes are the squares among the classes prime to 840, and Mordell observed that no identity of this kind can settle a square class.[1] The conjecture has been verified for all $n \leq 10^{17}$ by Salez [22] and for all $n \leq 10^{18}$ by Mihnea and Dumitru [16]. It remains open.

Let $\mathscr{E}(N)$ be the set of $n \leq N$ for which (1) has no solution. Vaughan [24] proved in 1970 that

$$|\mathscr{E}(N)| \ll N \exp\bigl(-c(\log N)^{2/3}\bigr) \tag{2}$$

for some constant $c > 0$, by applying the large sieve to the residue classes produced by one family of identities. Elsholtz and Tao [10] later determined the average number of solutions of (1) for prime n. Pomerance and Weingartner [20] extended (2) to m/n uniformly in m, with the exponent $C(\log N)^{2/3}/\varphi(m)^{1/3}$ for an absolute constant C. In both

[1] An identity that solves (1) for all n in a class r modulo q exists only if r is not a square modulo q; see [10, Section 2] for the principle and the families of identities behind it. The same computation decides a recent claim of a proof for the primes $p \equiv 1 \pmod 4$ [8]. It reduces the problem to a finite family of classes $p \equiv -4\alpha s^2 \pmod M$ with $M = 4\alpha sr - 1$ that is supposed to cover every prime. Since $M \equiv 3 \pmod 4$, $M \equiv -1$ modulo every odd prime factor of α, and $M \equiv -1 \pmod 8$ when α is even, quadratic reciprocity gives the Jacobi symbol $\left(\frac{-4\alpha s^2}{M}\right) = -1$, so no such class contains a square, and the class $p \equiv 1$ modulo 4 times the product of the moduli is never covered. By Dirichlet's theorem it contains infinitely many primes, all $\equiv 1 \pmod 4$. The reduction is therefore conditional, as its author states.

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papers the number of residue classes at each prime is controlled only up to constant factors, and no numerical value of c follows.[2] Dahan [6] has recently run Vaughan's large-sieve scheme on a two-parameter Type II system for the primes $n \equiv 1 \pmod{24}$, which contain every prime in $\mathscr{E}(N)$, and obtained (2) for these primes with $c = 0.065$. The sum that governs his system has the leading constant $9/(2\pi^2)$, found by Liu [13] and, with a power-saving error term, in [4]; with it the same argument gives $c = 0.0943$ for the primes. Here we give an explicit constant for the whole set $\mathscr{E}(N)$, and we take it beyond the value that the residue classes reachable by the Bombieri–Vinogradov theorem can give.

Theorem A. *For every $\varepsilon > 0$ there is $N_0(\varepsilon)$ such that, for all $N \ge N_0(\varepsilon)$,*

$$|\mathscr{E}(N)| \le N \exp\bigl(-(c_1-\varepsilon)(\log N)^{2/3}\bigr), \tag{3}$$

where

$$c_1 = \frac{3}{2}\Bigl(\frac{9}{8\pi^2}\Bigl(\frac{40}{49} - 6\mathscr{J}\Bigr)\Bigr)^{1/3} \tag{4}$$

and $\mathscr{J}$ is the integral (27) of Section VI. Moreover $0 \le \mathscr{J} \le 533/64680$, so that

$$c_1 \ge c_{\mathrm{var}} = \Bigl(\frac{286983}{98560\,\pi^2}\Bigr)^{1/3} = 0.6657102497\ldots \tag{5}$$

The number $40/49$ is the share of the triangle of factorisations (Figure 1) covered by the regions whose residue classes we count, and $6\mathscr{J}$ is the part of that share which our count misses. Numerical integration gives $\mathscr{J} = 0.0003$ and $c_1 = 0.6792$ (Section IX); two independent programs agree on c_1 to four decimals, and a Monte Carlo simulation agrees with the integrand. The lower bound (5) is proved in closed form.

The second theorem concerns the classes that the Bombieri–Vinogradov theorem can count. For a prime $p \equiv 3 \pmod 4$ with $P = (p+1)/4$, let $\Omega_p^\sharp$ be the set of classes $-u/a \pmod p$ attached to factorisations $P = ua\alpha$ with $\gcd(u,a) = 1$ in which one of the three products ua, $u\alpha$, $a\alpha$ is at most $\sqrt{p}$ (Section II). Let $L^\sharp(Q)$ be the sum that the large sieve attaches to these classes, as in (28) below.

Theorem B. *For every $\varepsilon > 0$ there is $N_1(\varepsilon)$ such that, for all $N \ge N_1(\varepsilon)$ and all $Q \ge 1$,*

$$\frac{N+Q^2}{L^\sharp(Q)} \ge N \exp\bigl(-(c_0+\varepsilon)(\log N)^{2/3}\bigr),$$

where

$$c_0 = \frac{9}{2^{8/3}\pi^{2/3}} = \frac{3}{2}\Bigl(\frac{27}{32\pi^2}\Bigr)^{1/3} = 0.6607886973\ldots \tag{6}$$

The classes $\Omega_p^\sharp$ come from the three corners of the triangle, and with them the large sieve gives exactly the constant c_0: Theorem B is the upper bound, and the corners alone already give c_0 (Corollary V.3). The number $27/(32\pi^2)$ inside (6) is the density of these classes. The gain from c_0 to c_1 comes from the classes of the centre of the triangle, which lie beyond the reach of the Bombieri–Vinogradov theorem.

The method. Each prime $p \equiv 3 \pmod 4$ carries the identities of Type II [10]: for every factorisation $P = ua\alpha$ with $\gcd(u,a) = 1$, every $n \equiv -u/a \pmod p$ has a solution of (1) (Lemma II.2). In logarithmic coordinates these factorisations fill a triangle (Figure 1). To sieve with their classes one must count them over the primes and show that they are distinct. Counting requires primes in progressions to the moduli $4ua$, $4u\alpha$ or $4a\alpha$. The Bombieri–Vinogradov theorem controls such moduli up to about $\sqrt{p}$, which cuts out three corners of the triangle, where one product of two variables is at most $\sqrt{p}$. In each corner the class has a normal form of small height, u/a, $4u^2\alpha$ or $(4a^2\alpha)^{-1}$, which separates the classes inside that corner, and two arguments with explicit inequalities show that the corner A shares no class with B or C. The corners B and C do share classes, but only along a family described by six integers, which contributes a convergent series (Proposition III.2). The corners cover three quarters of the triangle and give c_0.

The rest of the triangle is the centre, where u, a and α are all below $\sqrt{P}$. There the class u/a is a Farey fraction of order $\sqrt{P}$, so the classes of the centre are distinct. They differ from the classes of the corners, apart from a family described by five integers, which contains one explicit polynomial family of primes and again contributes a convergent series (Propositions IV.3 and IV.4). To count the centre we use a theorem of Maynard [15]: the primes $p \equiv -1 \pmod q$ have the expected number for almost all moduli q up to $p^{1/2+\vartheta}$ with $\vartheta < 1/42$, provided that q has a divisor in a range that depends on ϑ. A divisor chain (Lemma VI.3) finds such a divisor for most moduli in three strips of the centre along the corners, and we compute the proportion it misses (Lemma VI.4). If nothing were missed, the strips would add $6(\frac{1}{84} - \frac{3}{3528}) = \frac{13}{196}$ to the share $\frac{3}{4}$ of the corners; the loss $6\mathscr{J}$ is 0.0018.

The corners, their normal forms, the treatment of the centre and the rule that finds the divisors were selected by the engine Anaxia of the BLLM project [1, 2].[3]

The rest of the argument is analytic. A Rankin-type upper bound and a tilted lower bound determine the sum of the large sieve to leading order once the density of the classes is

[2] In [20, Section 4] the number of classes, summed over the primes $p \le x$ with weight $1/p$, is shown to be of order $(\log x)^2$: the upper bound comes from the Brun–Titchmarsh inequality and the lower bound from the Bombieri–Vinogradov theorem, with different implied constants, and the exponent is written through these two constants.

[3] Anaxia, version 3.0.5, builds constructions by typed crossing of proof components and tests them on data. We ran it twice. On the corners it tested 128 constructions on the 39,322 primes $p \equiv 3 \pmod 4$ below 10^6 and kept the seven built from one, two or three corners with matching normal forms; the best of them gives c_0. On the centre it tested 37 constructions and accepted five, the best of which is the construction of this paper. It rejected the forms $4u^2\alpha$ and $(4a^2\alpha)^{-1}$ in the centre, and a rule that omits the bound on the steps of the divisor chain, by finding counterexamples on data. Section XI describes both runs in full. The engine selected the constructions; the proofs are ours.

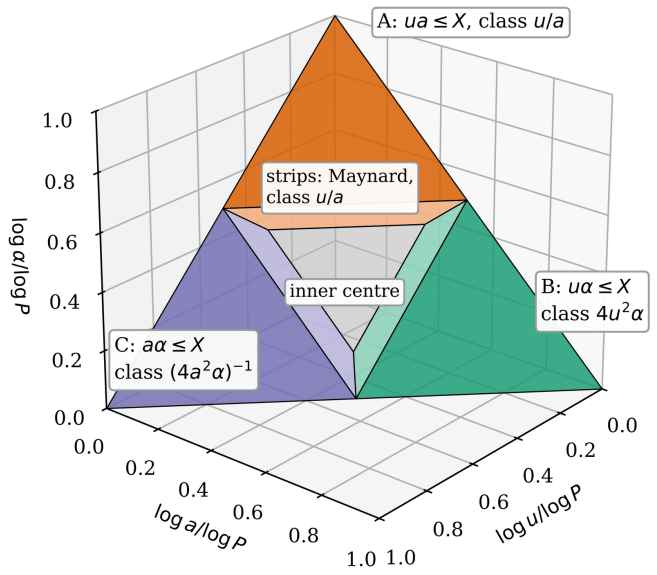


FIG. 1. In the coordinates $(\log u, \log a, \log \alpha)/\log P$ the factorisations $P = ua\alpha$ fill a triangle. The three corners A, B, C are the parts in which one product of two variables is at most X; each carries its own normal form of the class u/a (Lemma II.3). At $X = \sqrt{p}$ each corner has a quarter of the area. In the centre the three coordinates are below $\frac{1}{2}$ and the class u/a itself separates the classes (Proposition IV.3). The light strips of the centre along the corners, of depth $\frac{1}{42}$, are counted with Maynard's theorem (Section VI); they are drawn wider than they are. The inner centre would need moduli beyond Maynard's range.

known (Proposition VII.2); the same upper bound proves Theorem B. Section IX computes the constant, Section X compares regions of the triangle (Table I) and explains where the method stops, and Appendix A contains every computation in full.

II. TYPE II TRIPLES AND THEIR CLASSES

Throughout, p denotes a prime with $p \equiv 3 \pmod 4$, and $P = (p+1)/4$, so that $p = 4P-1$ and $4P \equiv 1 \pmod p$.

Definition II.1. A *Type II triple* for p is $t = (u,a,\alpha) \in \mathbb{N}^3$ with $ua\alpha = P$ and $\gcd(u,a) = 1$. Their set is $\mathscr{T}(p)$. The *class* of t is $\mathrm{c}(t) = ua^{-1} \bmod p$, which is defined because $1 \le a \le P < p$.

Lemma II.2. *Let $t = (u,a,\alpha) \in \mathscr{T}(p)$ and $n \ge 1$ with $n \equiv -\mathrm{c}(t) \pmod p$. Then $v = (an+u)/p$ is a positive integer and*

$$\frac{4}{n} = \frac{1}{\alpha uv} + \frac{1}{a\alpha nu} + \frac{1}{a\alpha nv}. \tag{7}$$

Proof. The congruence $n \equiv -u/a$ means $p \mid an+u$, and $an+u \ge 2$, so v is a positive integer. Then $an+u = pv = (4ua\alpha - 1)v$, that is $an+u+v = 4a\alpha uv$. Dividing by $a\alpha nuv$ gives (7). □

When n is prime and $n \nmid uv$, two of the three denominators in (7) are divisible by n and the third is not; these are the solutions of Type II in the sense of [10].[4] By Lemma II.2, an integer $n \in \mathscr{E}(N)$ lies, for every prime $p \equiv 3 \pmod 4$, outside every class $-\mathrm{c}(t)$ with $t \in \mathscr{T}(p)$. This is the input to the large sieve.

Lemma II.3. *For every $t = (u,a,\alpha) \in \mathscr{T}(p)$,*

$$\mathrm{c}(t) \equiv 4u^2\alpha \equiv (4a^2\alpha)^{-1} \pmod p. \tag{8}$$

Proof. From $p = 4ua\alpha - 1$ we get the identities

$$4(u^2\alpha)a - u = up, \qquad 4(a^2\alpha)u - a = ap.$$

The first gives $4u^2\alpha \cdot a \equiv u$, so $u/a \equiv 4u^2\alpha$. The second gives $4a^2\alpha \cdot u \equiv a$, so $u/a \equiv (4a^2\alpha)^{-1}$. □

We call u/a, $4u^2\alpha$ and $(4a^2\alpha)^{-1}$ the three normal forms of the class. Each is useful where its height is small.

Definition II.4. For $X \ge 1$ the three *corners* at level X are

$$A_p(X) = \{t \in \mathscr{T}(p) : u \ge 2,\ a \ge 2,\ ua \le X\},$$
$$B_p(X) = \{t \in \mathscr{T}(p) : u\alpha \le X\},$$
$$C_p(X) = \{t \in \mathscr{T}(p) : a\alpha \le X\}.$$

The corner A omits the triples with $u = 1$ or $a = 1$, which are needed in the proof of Proposition III.1 and are too few to matter in the count (they contribute $O(\log X)$ to the sums of Section V). The map $\iota(u,a,\alpha) = (a,u,\alpha)$ is an involution of $\mathscr{T}(p)$ with $\mathrm{c}(\iota t) = \mathrm{c}(t)^{-1}$; it maps $A_p(X)$ onto itself and exchanges $B_p(X)$ and $C_p(X)$. In particular $|B_p(X)| = |C_p(X)|$.

Lemma II.5. *Suppose $16X^2 \le p$. Then:*

(i) the corners $A_p(X)$, $B_p(X)$, $C_p(X)$ are pairwise disjoint;

(ii) on $A_p(X)$ one has $u,a \le X$; on $B_p(X)$ one has $u^2\alpha \le X^2$; on $C_p(X)$ one has $a^2\alpha \le X^2$.

Proof. (i) Two of the three conditions force the product of the two products to be at most X^2. For A and B this is $(ua)(u\alpha) = uP \le X^2$; for A and C it is $aP \le X^2$; for B and C it is $\alpha P \le X^2$. In each case $P \le X^2 \le p/16 < P$, which is impossible. (ii) On A, u and a divide $ua \le X$. On B, $u^2\alpha = u \cdot u\alpha \le (u\alpha)^2 \le X^2$, and similarly on C. □

III. DISTINCT CLASSES IN THE CORNERS

This section proves that the classes of the three corners are distinct, apart from a sparse set of coincidences between B and C.

Proposition III.1. *Suppose $16X^2 \le p$. Then c is injective on each of $A_p(X)$, $B_p(X)$ and $C_p(X)$, and no class of $A_p(X)$ is a class of $B_p(X)$ or of $C_p(X)$.*

[4] The identity (7) with $p = 4ua\alpha - 1$ is the one behind Vaughan's argument [24] and the Type II search of [6]; in [4] we used it with α replaced by the witness s of that search.

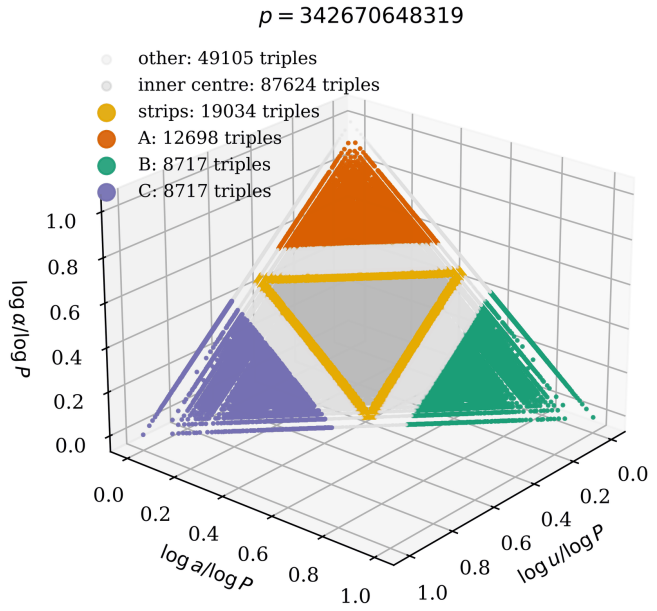


FIG. 2. The 185,895 Type II triples of the prime $p = 342670648319$, for which $P = 2^8 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 13 \cdot 17 \cdot 19 \cdot 23$, with the corners at level $X = \sqrt{p}/16$, the strips of the centre (largest coordinate at least $P^{1/2-1/42}$) and the inner centre. The light grey triples lie in no corner and outside the centre. The corners B and C have the same size, as the involution ι shows.

Proof. Inside A. Let $t,t' \in A_p(X)$ with $u/a \equiv u'/a'$. Then $p \mid ua' - u'a$, and $|ua' - u'a| < X^2 < p$ by Lemma II.5(ii), so $ua' = u'a$. Since $\gcd(u,a) = \gcd(u',a') = 1$, this gives $u = u'$, $a = a'$ and then $\alpha = \alpha'$.

Inside B. Let $t,t' \in B_p(X)$ with equal classes. By (8), $4u^2\alpha \equiv 4u'^2\alpha'$, and both sides lie in $[1,4X^2] \subset [1,p)$, so $h = u^2\alpha$ equals $h' = u'^2\alpha'$. Now $ha = u \cdot ua\alpha = uP$, so $u/a = h/P$ as rational numbers; since $\gcd(u,a) = 1$, the pair (u,a) is the reduced form of h/P. The same holds for t', so $t = t'$.

Inside C. The involution ι maps C onto B and inverts classes, so injectivity on C follows from injectivity on B.

A against B. Let $t = (u,a,\alpha) \in A_p(X)$ and $t' = (u',a',\alpha') \in B_p(X)$ have the same class, and put $h = u'^2\alpha'$, so that $u/a \equiv 4h$ by (8). Then $4ah - u = wp$ for some integer w.

If $w \le -1$ then $4ah \le u - p < 0$, which is impossible. If $w = 0$ then $4ah = u$, so $a \mid u$, and $\gcd(u,a) = 1$ forces $a = 1$, which is excluded in A. Hence $w \ge 1$. Substituting $p = 4ua\alpha - 1$ in $4ah = u + wp$ gives

$$4ah = u - w + 4ua\alpha w,$$

so $4a \mid u - w$. Write $u - w = 4aj$ with $j \in \mathbb{Z}$; then

$$h = j + wu\alpha, \qquad w = u - 4aj. \tag{9}$$

Case $j \le -1$. Then $w = u + 4a|j|$ and

$$h = -|j| + (u + 4a|j|)u\alpha = |j|p + u^2\alpha > p,$$

while $h \le X^2 < p$ by Lemma II.5(ii). This is impossible.

Case $j = 0$. Then $h = u^2\alpha$, and, as in the proof for B, $u'/a' = h/P = u^2\alpha/(ua\alpha) = u/a$, so $t' = t$. But A and B are disjoint by Lemma II.5(i).

Case $j \ge 1$. Since $w \ge 1$, (9) gives $4aj \le u - 1$, so $aj < u/4$. Put $d = u'\alpha'$. Then $P = a'd$ and $h = u'd$, and from (9),

$$ah = aj + wua\alpha = aj + wP = aj + wa'd,$$

that is

$$d\,(au' - a'w) = aj. \tag{10}$$

The right side is positive, so $au' - a'w \ge 1$ and $d \le aj < u/4$. Hence $h = u'^2\alpha' \le (u'\alpha')^2 = d^2 < u^2/16$, and

$$p < u + wp = 4ah < \frac{au^2}{4} \le \frac{(ua)^2}{4} \le \frac{X^2}{4} < p,$$

a contradiction. So no class of A is a class of B.

A against C. If $t \in A_p(X)$ and $t' \in C_p(X)$ had the same class, then $\iota t \in A_p(X)$ and $\iota t' \in B_p(X)$ would have the same class $\mathrm{c}(t)^{-1}$, which the previous step excludes. □

The corners B and C can share a class. The next proposition describes every such coincidence.

Proposition III.2. *Suppose $16X^2 \le p$, and let $t = (u,a,\alpha) \in B_p(X)$ and $t' = (u',a',\alpha') \in C_p(X)$ have the same class. Put*

$$d = u\alpha, \quad e = a'\alpha', \quad M = ua', \quad k = \alpha\alpha',$$

$G = \gcd(d,e)$, $d = Gd_1$ and $e = Ge_1$. Then there are integers $r,m,\tau \ge 1$ such that

$$P = Gd_1e_1(4Gr + \tau), \tag{11}$$

$$M = d_1e_1\tau(4Gr + \tau) + r, \tag{12}$$

$$G = 4k\tau r + m, \tag{13}$$

$$m(4k\tau r + m) = k\tau^2 + \frac{kr}{d_1e_1}, \quad d_1e_1 \mid kr, \tag{14}$$

and

$$\tau \ge 4mr, \qquad G \ge 16kmr^2, \qquad P \ge 1024\,k^2m^2r^5d_1e_1. \tag{15}$$

The integers k,m,r,d_1,e_1,τ and u determine the pair (t,t').

Proof. Step 1: the integer r. By (8), $4u^2\alpha \equiv (4a'^2\alpha')^{-1}$, so $16u^2\alpha a'^2\alpha' \equiv 1 \equiv 16P^2$ and therefore $u^2\alpha a'^2\alpha' \equiv P^2$, that is, $deM \equiv P^2 \pmod p$. On the other hand $P = ad = u'e$, so $P^2 = de \cdot au'$. Since $p \nmid de$, we get $au' \equiv M \pmod p$; write $au' = M + rp$ with $r \in \mathbb{Z}$, so that

$$P^2 = de\,(M + rp). \tag{16}$$

If $r = 0$ then $au' = ua'$, and since both pairs are coprime, $u = u'$ and $a = a'$, so $t = t'$, which contradicts Lemma II.5(i). If $r \le -1$ then $M = au' + |r|p > p$, but $M = ua' \le (u\alpha)(a'\alpha') \le X^2 < p$. Hence $r \ge 1$.

Step 2: the integer $s = P - 4der$. Expanding (16) with $p = 4P - 1$ gives

$$Ps = de\,(M - r), \qquad s = P - 4der. \tag{17}$$

Since $M \geq 1$, (16) gives $P^2 > der(4P-1)$, hence $4der < P^2/(P-\frac{1}{4}) < P+1$, so $s \geq 0$. If $s=0$ then $M=r$ and $P=4der$; then $a=P/d=4er=4eua'$, so $u \mid a$ and $u=1$; similarly $u'=P/e=4dr=4dua'$, so $a' \mid u'$ and $a'=1$. Then $r=M=1$ and $P=4de \leq 4X^2 \leq p/4 < P$, a contradiction. Hence $s \geq 1$.

Step 3: the parametrisation. Both d and e divide P and $4der$, hence s, so the least common multiple Gd_1e_1 divides s. Write $s = Gd_1e_1\tau$ with $\tau \geq 1$. Then $P = s+4der = Gd_1e_1(4Gr+\tau)$, which is (11). Inserting this into (17) and cancelling $G^2d_1e_1$ gives (12). Next, $de = ua' \cdot \alpha\alpha' = kM$, so $kM = G^2d_1e_1$. With (12) this reads $kd_1e_1\tau(4Gr+\tau)+kr = G^2d_1e_1$, so $d_1e_1 \mid kr$ and

$$G^2 - k\tau(4Gr+\tau) = \frac{kr}{d_1e_1} \geq 1. \tag{18}$$

Hence $G(G-4k\tau r) > k\tau^2 > 0$, so $m = G-4k\tau r \geq 1$, which is (13), and (18) becomes (14).

Step 4: the inequalities. From (14), $4k\tau rm+m^2 \leq k\tau^2+kr$. Suppose $\tau < 4mr$. The integers τ and $4mr-\tau$ are positive with sum $4mr$, so their product is at least $4mr-1 \geq 3r \geq r$. Then $k\tau(4mr-\tau) \geq kr$, while the inequality above gives $k\tau(4mr-\tau) \leq kr-m^2 < kr$. This contradiction proves $\tau \geq 4mr$. Then $G = 4k\tau r+m \geq 16kmr^2$, and from (11),

$$P \geq Gd_1e_1 \cdot 4Gr = 4rG^2d_1e_1 \geq 1024k^2m^2r^5d_1e_1.$$

Step 5: recovery of the pair. Given k,m,r,d_1,e_1,τ, the integers G, P, d, e and M are fixed by (13), (11) and (12). Given also u, we recover $\alpha = d/u$, $a' = M/u$, $\alpha' = e/a'$, $a = P/d$ and $u' = P/e$. □

For a prime $p \equiv 3 \pmod 4$ let $\nu(p)$ be the number of pairs (t,t') with $t \in B_p(\sqrt{p}/4)$, $t' \in C_p(\sqrt{p}/4)$ and $\mathrm{c}(t) = \mathrm{c}(t')$. For every $X \leq \sqrt{p}/4$ the corners at level X lie inside those at level $\sqrt{p}/4$, so the number of coincidences between $B_p(X)$ and $C_p(X)$ is at most $\nu(p)$.

Corollary III.3. $\sum_p \nu(p)/p < \infty$.

Proof. Fix $\eta \in (0,\frac{1}{4})$. By Proposition III.2, each coincidence at p gives integers $k,m,r,d_1,e_1,\tau \geq 1$ that fix P, together with a divisor u of $M < p$; there are at most $C_\eta P^\eta$ such divisors. For fixed k,m,r,d_1,e_1, the equation (14) is a quadratic equation in τ with leading coefficient $k \neq 0$, so it has at most two solutions. The pairs (d_1,e_1) with $d_1e_1 \mid kr$ number at most $\tau_3(kr) \leq C_\eta(kr)^\eta$, where τ_3 counts ordered factorisations into three factors. Using $p > P \geq 1024k^2m^2r^5$, we obtain

$$\sum_p \frac{\nu(p)}{p} \leq C'_\eta \sum_{k,m,r\geq 1} \frac{(kr)^\eta}{(k^2m^2r^5)^{1-\eta}},$$

and the series converges because the exponents of k, m and r are $2-3\eta$, $2-2\eta$ and $5-6\eta$, all greater than 1. □

IV. THE CENTRE

Definition IV.1. The *centre* of $\mathscr{T}(p)$ is

$$\mathscr{Z}_p = \{t \in \mathscr{T}(p) : u^2 < P,\ a^2 < P,\ \alpha^2 < P\}.$$

Lemma IV.2. *Let $t=(u,a,\alpha) \in \mathscr{Z}_p$. Then $u,a,\alpha \geq 2$, $u \neq a$, and each of the products ua, $u\alpha$, $a\alpha$ exceeds $\sqrt{P}$. In particular $\mathscr{Z}_p$ is disjoint from the corners at every level $X \leq \sqrt{p}/2$, and ι maps $\mathscr{Z}_p$ onto itself.*

Proof. Since $a\alpha < \sqrt{P}\sqrt{P} = P$, we have $u = P/(a\alpha) > 1$, and in the same way $a,\alpha \geq 2$. Then $u \neq a$ because $\gcd(u,a)=1$. Next $ua = P/\alpha > P/\sqrt{P} = \sqrt{P}$, and in the same way for the other products. Finally $\sqrt{P} > \sqrt{p}/2$, and the conditions of the centre are symmetric in u and a. □

In the centre the first normal form u/a has numerator and denominator below $\sqrt{P}$, as for a Farey fraction of order $\sqrt{P}$, and this separates the classes.

Proposition IV.3. *Let $X \leq \sqrt{p}/4$.*

(i) c *is injective on $\mathscr{Z}_p$, and no class of $\mathscr{Z}_p$ is a class of $A_p(X)$.*

(ii) No class of a triple of $\mathscr{Z}_p$ with $u<a$ is a class of $B_p(X)$, and no class of a triple of $\mathscr{Z}_p$ with $u>a$ is a class of $C_p(X)$.

Proof. (i) Let $t,t' \in \mathscr{Z}_p$ have the same class, so that $p \mid ua'-u'a$. From $u^2 < P$ and $a'^2 < P$ we get $(ua')^2 < P^2$, so $ua' < P$, and in the same way $u'a < P$. Hence $|ua'-u'a| < P < p$, so $ua' = u'a$, and as before $t=t'$. If instead $t' \in A_p(X)$, then $u',a' \leq X$ by Lemma II.5(ii), so ua' and $u'a$ are below $\sqrt{P}\cdot\sqrt{p}/4 < p$; again $ua' = u'a$ and $t=t'$, which contradicts Lemma IV.2.

(ii) Let $t \in \mathscr{Z}_p$ with $u<a$. Then $4u^2\alpha < 4ua\alpha = p+1$, and $4u^2\alpha \neq p$ because p is odd, so $1 \leq 4u^2\alpha < p$. If $t' \in B_p(X)$ has the same class, then $1 \leq 4u'^2\alpha' \leq 4X^2 < p$ by Lemma II.5(ii), and (8) gives $4u^2\alpha = 4u'^2\alpha'$. As in the proof of Proposition III.1, both (u,a) and (u',a') are the reduced form of $u^2\alpha/P$, so $t=t'$, which contradicts Lemma IV.2. The second statement follows by applying ι, which maps the triples of $\mathscr{Z}_p$ with $u>a$ to those with $u<a$, maps $C_p(X)$ onto $B_p(X)$, and inverts classes. □

The remaining coincidences, between the centre with $u<a$ and the corner C (and their images under ι), have a parametrisation by five integers.

Proposition IV.4. *Let $X \leq \sqrt{p}/4$, and let $t=(u,a,\alpha) \in \mathscr{Z}_p$ with $u<a$ and $t''=(u'',a'',\alpha'') \in C_p(X)$ have the same class. Put $s=a''\alpha''$. Then there are integers $k,D,j \geq 1$ such that*

$$u''a - ua'' = kp, \tag{19}$$

$$a = j+4ks, \tag{20}$$

$$uD = sk, \tag{21}$$

$$\alpha j(j+4ks) = sa'' - D, \tag{22}$$

and $4ks < a < \sqrt{P}$. The integers a'', α'', k, D and j determine the pair (t,t'').

Proof. The equality of classes gives $p \mid u''a-ua''$; write $u''a-ua'' = kp$. If $k=0$ then $u''a = ua''$ and $t=t''$, which contradicts Lemma IV.2. If $k \leq -1$ then $ua'' \geq p+u''a > p$,

while $ua'' < \sqrt{P}\cdot X < p$. Hence $k \geq 1$. Multiplying (19) by s and using $u''s = P$ and $p = 4P-1$ gives

$$P(a-4ks) = s(ua''-k).$$

From $k(4P-1) < u''a$ we get $ks(4P-1) < Pa < P(4P-1)$, so $ks < P$. If $4ks > a$, the left side is at most $-P < -sk \leq s(ua''-k)$, which is impossible. So $j = a-4ks \geq 0$. If $j = 0$ then $ua'' = k$, so $a = 4ks = 4ua''s$ is a multiple of u, and $\gcd(u,a) = 1$ forces $u = 1$, which contradicts Lemma IV.2. Hence $j \geq 1$, which is (20). Inserting $P = ua\alpha$,

$$ua\alpha j = s(ua''-k), \qquad \text{so} \qquad u(sa''-a\alpha j) = sk.$$

Put $D = sa''-a\alpha j$; then $uD = sk \geq 1$, so $D \geq 1$, which is (21), and $\alpha ja = sa''-D$ is (22). Finally $a < \sqrt{P}$ because $t \in \mathscr{Z}_p$, and $4ks = a-j < a$. Given a'', α'', k, D and j, we recover $s = a''\alpha''$, $u = sk/D$, $a = j+4ks$, $\alpha = (sa''-D)/(ja)$, $P = ua\alpha$ and $u'' = P/s$. □

For a prime $p \equiv 3 \pmod 4$ let $\nu'(p)$ be the number of pairs (t,t'') with $t \in \mathscr{Z}_p$, $t'' \in B_p(\sqrt{p}/4) \cup C_p(\sqrt{p}/4)$ and $\mathrm{c}(t) = \mathrm{c}(t'')$.

Corollary IV.5. $\sum_p \nu'(p)/p < \infty$.

Proof. By Proposition IV.3(ii), a pair counted by $\nu'(p)$ has either $u < a$ and $t'' \in C_p(\sqrt{p}/4)$, or $u > a$ and $t'' \in B_p(\sqrt{p}/4)$, and ι exchanges the two kinds. It is therefore enough to count the first kind. By Proposition IV.4, a pair of the first kind at a prime $p \leq y$ gives $s = a''\alpha''$ and k with $ks < \sqrt{P}/4 < \sqrt{y}$, a divisor D of sk, and a divisor j of $sa''-D$, which lies in $[1,s^2] \subset [1,y]$. Fix $\eta > 0$. For given (s,k) there are at most $\tau(s)\tau(sk)\max_{m\leq y}\tau(m) \ll_\eta y^\eta$ choices of (a'',α'',D,j), and the pairs (s,k) with $ks < \sqrt{y}$ number $O(\sqrt{y}\log y)$. Hence the pairs of the first kind at the primes $p \leq y$ number $O_\eta(y^{1/2+2\eta})$. With $\eta = \frac{1}{8}$, partial summation gives $\sum_{p\leq y}\nu'(p)/p = O(1)$. □

Remark IV.6 (A family of coincidences)**.** For an integer $\alpha \geq 1$ put

$$s = 16\alpha+1, \quad a = 8(8\alpha+1), \quad P = sa\alpha, \quad p = 4P-1.$$

When p is prime, $t = (s,a,\alpha)$ and $t'' = (a\alpha,s,1)$ are Type II triples for p with the same class, since

$$(a\alpha)a - s\cdot s = 4P-1 = p.$$

They are coprime pairs because s is odd with $2(8\alpha+1)-s = 1$, and $\gcd(s,\alpha) = 1$. This is the case $(k,D,j) = (1,1,4)$ of Proposition IV.4, with $a'' = s$ and $\alpha'' = 1$. The triple t lies in the centre exactly when $a^2 < P$, that is $16\alpha^2-63\alpha-8 > 0$, that is $\alpha \geq 5$, and t'' lies in $C_p(\sqrt{p}/4)$ because $s < 16\alpha^{3/2} < \sqrt{p}/4$. The value $\alpha = 5$ gives the prime $p = 531359$, with $t = (81,328,5)$ and $t'' = (1640,81,1)$:

$$81^2 - 328\cdot 1640 = -531359.$$

The value $\alpha = 7$ gives the prime $p = 1442783$, with $t = (113,456,7)$ and $t'' = (3192,113,1)$. Together with their images under ι, these four pairs are all the pairs counted by $\nu'(p)$ for the primes $p \equiv 3 \pmod 4$ below 10^7. Since p is a cubic polynomial in α, the family by itself contributes a convergent series to $\sum_p \nu'(p)/p$.

V. COUNTING THE CLASSES OF THE CORNERS

A. Mean values

Let $\kappa = 9/(2\pi^2) = 0.4559453\ldots$, the constant of [4, 13]. For $X \geq 1$ define

$$S_A(X) = \sum_{\substack{u,a\geq 2,\ ua\leq X\\ \gcd(u,a)=1}} \frac{1}{\varphi(4ua)},$$

$$S_B(X) = \sum_{u\alpha\leq X} \frac{\varphi(u)}{u\,\varphi(4u\alpha)}.$$

The weight $1/\varphi(4ua)$ is the density of the primes $p \equiv -1 \pmod{4ua}$, that is, of the primes for which (u,a,α) with $\alpha = P/(ua)$ is a triple of $A_p(X)$. In S_B the extra factor $\varphi(u)/u$ is the density of the condition $\gcd(u,a) = 1$ among the primes $p \equiv -1 \pmod{4u\alpha}$.[5]

Lemma V.1. *As $X \to \infty$,*

$$S_A(X) = \frac{\kappa}{2}\log^2 X + O(\log X),$$
$$S_B(X) = \frac{\kappa}{2}\log^2 X + O(\log X).$$

Sketch. Since $\varphi(4n) = 2\varphi(n)$ for odd n and $4\varphi(n)$ for even n, we have $1/\varphi(4n) = \psi(n)/(2n)$ with $\psi(n) = \prod_{q\mid n,\, q \text{ odd}} q/(q-1)$. Thus $S_B(X) = \frac{1}{2}\sum_{u\alpha\leq X} W_B(u,\alpha)/(u\alpha)$ with $W_B(u,\alpha) = \psi(u\alpha)\varphi(u)/u$, a function that is multiplicative in the pair. Writing $W_B(u,\alpha) = \sum_{d\mid u,e\mid\alpha} g(d,e)$, the function g is supported on $g(1,1) = 1$, $g(1,q) = 1/(q-1)$ and $g(q,q) = -1/(q-1)$ for odd primes q, and $g(2,1) = -\frac{1}{2}$. Then

$$S_B(X) = \frac{1}{2}\sum_{d,e}\frac{g(d,e)}{de}H\Big(\frac{X}{de}\Big), \qquad H(Y) = \sum_{mn\leq Y}\frac{1}{mn},$$

and $H(Y) = \frac{1}{2}\log^2 Y + 2\gamma\log Y + O(1)$ for $Y \geq 1$. The sum $\sum_{d,e} g(d,e)/(de)$ is the Euler product of the local factors $\frac{3}{4}$ at 2 and $1+q^{-2}$ at odd q, that is $\frac{3}{4}\cdot\frac{12}{\pi^2} = \frac{9}{\pi^2} = 2\kappa$, and the main term is $\frac{1}{2}\cdot 2\kappa\cdot\frac{1}{2}\log^2 X$. The sum S_A is treated in the same way with $W_A(u,a) = \psi(ua)\mathbf{1}[\gcd(u,a) = 1]$; its local factors are again $\frac{3}{4}$ and $1+q^{-2}$, and the triples with $u = 1$ or $a = 1$ contribute $O(\log X)$. The details are in Appendices A 1 and A 2. □

The two corners have the same constant because they have the same local factors (Table II in Appendix A 1).

[5] For a prime $q \mid u$, the primes $p \equiv -1 \pmod{4u\alpha}$ fall into q classes modulo $4u\alpha q$, all prime to $4u\alpha q$, and $q \mid a = (p+1)/(4u\alpha)$ holds on exactly one of them. By inclusion and exclusion over the prime factors of u, the condition $\gcd(u,a) = 1$ has density $\prod_{q\mid u}(1-1/q) = \varphi(u)/u$, because $\varphi(4u\alpha q) = q\,\varphi(4u\alpha)$ when $q \mid u$.

B. Primes

To count the corners over the primes we use dyadic blocks. Fix $\delta \in (0, \frac{1}{10}]$. For $j \ge 1$ put

$$I_j = (2^j, 2^{j+1}], \qquad X_j = 2^{j(1/2-\delta)}.$$

For $p \in I_j$ we have $16X_j^2 \le 16 \cdot 2^{j(1-2\delta)} \le p$ once $j \ge 2/\delta$. We write $|K_p(X)|$ for the size of a corner, with the convention $|K_p(X)| = 0$ when $p \not\equiv 3 \pmod 4$.

Proposition V.2. *For $K \in \{A, B, C\}$, as $j \to \infty$,*

$$\sum_{p \in I_j} \frac{|K_p(X_j)|}{p} = \frac{\kappa}{2}\Big(\frac{1}{2} - \delta\Big)^2 (\log 2)^2\, j + O_\delta\big(j^{1/2}\big).$$

Proof. For C the statement is the one for B, since $|C_p(X)| = |B_p(X)|$. Let $x = 2^{j+1}$ and $z = j^{1/2}$.

Corner A. A pair (u, a) with $u, a \ge 2$, $\gcd(u, a) = 1$ and $ua \le X_j$ belongs to $A_p(X_j)$ exactly when $p \equiv -1 \pmod{4ua}$. Hence, for $y \in I_j$,

$$\sum_{2^j < p \le y} |A_p(X_j)| = \sum_{(u,a)} \big(\pi(y; 4ua, -1) - \pi(2^j; 4ua, -1)\big).$$

The moduli $4ua \le 4X_j$ are below $x^{1/2}(\log x)^{-B}$ for j large, and each modulus $4n$ arises from at most $\tau(n)$ pairs. The Bombieri–Vinogradov theorem [5], [7, Chapter 28] with these multiplicities (Appendix A 3) gives

$$\sum_{2^j < p \le y} |A_p(X_j)| = \big(\pi(y) - \pi(2^j)\big) S_A(X_j) + O\big(x(\log x)^{-3}\big)$$

uniformly in $y \in I_j$.

Corner B. Let $P_z = \prod_{q \le z} q$. A pair (u, α) with $u\alpha \le X_j$ belongs to $B_p(X_j)$ when $p \equiv -1 \pmod{4u\alpha}$ and $\gcd(u, a) = 1$, where $a = (p+1)/(4u\alpha)$. For a squarefree $d \mid u$ the condition $d \mid a$ is $p \equiv -1 \pmod{4u\alpha d}$. Inclusion and exclusion over the primes $q \le z$ that divide u count the pairs with no common prime factor $q \le z$:

$$|B_p^{(z)}| = \sum_{u\alpha \le X_j} \sum_{d \mid (u, P_z)} \mu(d)\, \mathbf{1}\big[p \equiv -1 \pmod{4u\alpha d}\big].$$

These moduli are at most $4X_j e^{2z}$, still in the range of the Bombieri–Vinogradov theorem, and a modulus $4n$ arises at most $\tau_3(n)$ times. The main term is $(\pi(y) - \pi(2^j)) S_B^{(z)}(X_j)$, where $S_B^{(z)}$ is S_B with $\varphi(u)/u$ replaced by $\prod_{q \mid u, q \le z}(1 - 1/q)$, and $S_B \le S_B^{(z)} \le S_B + O(\log^2 X_j / z)$ (Appendix A 5). Then $|B_p(X_j)| \le |B_p^{(z)}|$, and the difference counts pairs with $q \mid u$ and $q \mid a$ for some prime $q > z$. The Brun–Titchmarsh inequality [18] bounds their total over $p \in I_j$ by $O(\delta^{-1}\pi(I_j)\log^2 X_j / z)$ (Appendix A 5).

The weights $1/p$. Partial summation inside the block (Appendix A 8) turns the counts into

$$\sum_{p \in I_j} \frac{|K_p(X_j)|}{p} = S_K(X_j) \sum_{p \in I_j} \frac{1}{p} + O\Big(\frac{j}{\delta z}\Big) + O(1),$$

with $S_C = S_B$. By the prime number theorem with error term, $\sum_{p \in I_j} 1/p = 1/j + O(1/j^2)$. With $S_K(X_j) = \frac{\kappa}{2}(\frac{1}{2} - \delta)^2 (\log 2)^2 j^2 + O(j)$ from Lemma V.1 and $z = j^{1/2}$, this is the statement. □

For $p \in I_j$ with $j \ge 2/\delta$ let $\omega_\delta^{\mathrm{cor}}(p)$ be the number of distinct classes $-\mathrm{c}(t)$ with t in one of the three corners at level X_j, and let $\omega_\delta^{\mathrm{cor}}(p) = 0$ for all other primes. For a function ω on the primes define

$$N[\omega](\lambda) = \sum_{p \le e^\lambda} \frac{\omega(p)}{p}. \tag{23}$$

Corollary V.3. *As $\lambda \to \infty$, $N[\omega_\delta^{\mathrm{cor}}](\lambda) = (a_\delta^{\mathrm{cor}} + o(1))\lambda^2$ with*

$$a_\delta^{\mathrm{cor}} = \frac{3\kappa}{4}\Big(\frac{1}{2} - \delta\Big)^2.$$

In particular $a_\delta^{\mathrm{cor}} \to a_0 = 3\kappa/16 = 27/(32\pi^2)$ as $\delta \to 0$.

Proof. By Propositions III.1 and III.2,

$$\sum_K |K_p(X_j)| - \nu(p) \le \omega_\delta^{\mathrm{cor}}(p) \le \sum_K |K_p(X_j)|.$$

Summing Proposition V.2 over the three corners and over $2/\delta \le j \le J$ gives

$$\frac{3\kappa}{2}\Big(\frac{1}{2} - \delta\Big)^2 (\log 2)^2 \sum_{j \le J} j + O_\delta(J^{3/2}) = a_\delta^{\mathrm{cor}} (J \log 2)^2 + O_\delta(J^{3/2}).$$

The coincidences contribute $O(1)$ by Corollary III.3, and a partial block contributes $O(J)$. With $J \log 2 = \lambda + O(1)$ the corollary follows. □

VI. THE STRIPS OF THE CENTRE

A. Maynard's theorem

For $0 < \vartheta < \frac{1}{42}$ put

$$\theta(\vartheta) = \min\Big(\frac{1}{10} - \frac{7\vartheta}{5},\ \frac{1}{2} - 19\vartheta\Big). \tag{24}$$

Theorem VI.1 (Maynard [15, Corollary 1.2])**.** *Let $b \in \mathbb{Z}$, $0 < \vartheta < \frac{1}{42}$ and $0 < \eta < (1 - 42\vartheta)/4$, and let $Q_{\vartheta,\eta}(y)$ be the set of $q \le y^{1/2+\vartheta}$ that have a divisor d with $y^{2\vartheta+\eta} < d < y^{\theta(\vartheta)-\eta}$. Then for every $A > 0$,*

$$\sum_{\substack{q \in Q_{\vartheta,\eta}(y) \\ (q,b)=1}} \Big|\pi(y; q, b) - \frac{\pi(y)}{\varphi(q)}\Big| \ll_{b,\eta,A} \frac{y}{(\log y)^A}.$$

We use it with $b = -1$, so that every q is prime to b.[6] The theorem asks for a divisor of the modulus in a prescribed

[6] The range $(y^{2\vartheta}, y^{\theta(\vartheta)})$ for the divisor is not empty exactly when $\vartheta < \frac{1}{42}$, and its upper end is $y^{1/10 - 7\vartheta/5}$ exactly when $\vartheta \le \frac{1}{44}$. Its width is at least its lower end, $\theta(\vartheta) - 2\vartheta \ge 2\vartheta$, exactly when $\vartheta \le \frac{1}{54}$. These three values are where the integrand of (27) changes its shape (Figure 3).

range. For the moduli $4ua$ of the centre, the next lemma finds one from the factorisations of u and a.

B. A divisor in a window

Definition VI.2. Let $1 < A < B$ and $G = \min(B/A, A)$. For $n \geq 1$ let $T(n)$ be the product of the prime powers $q^e \parallel n$ with $q < G$, and let $q(n)$ be the largest prime factor of n in $[G, A]$, or $q(n) = 1$ if there is none. A pair (u, v) of positive integers *passes the rule* for the window (A, B) if

(R1) $u \in (A, B)$ or $v \in (A, B)$; or

(R2) uv has a prime factor in (A, B); or

(R3) $q(uv)\,T(uv) > A$.

Lemma VI.3. *If (u, v) passes the rule for (A, B), then uv has a divisor in (A, B).*

Proof. Under (R1) or (R2) this is clear. Suppose (R3) holds, and let $r_1, \dots, r_m$ be the prime factors of $T(uv)$, listed with multiplicity; each is below G. Put $d_0 = q(uv)$ and $d_i = d_0 r_1 \cdots r_i$. Since $q(uv) \geq G$ when it is a prime, d_0 is prime to $T(uv)$, so every d_i divides uv. Now $d_0 \leq A < d_m$, so there is a least $i \geq 1$ with $d_i > A$, and then

$$d_i = d_{i-1} r_i < A\,G \leq A \cdot \frac{B}{A} = B.$$

So d_i is a divisor of uv in (A, B). □

The bound $r_i < G \leq B/A$ on the steps of the chain is needed.[7] The condition $G \leq A$ keeps the primes of $T(uv)$ apart from $q(uv)$.

C. The density of the pairs that fail

Let $0 < \theta^- < \theta^+$ and put $\xi = \min(\theta^+ - \theta^-, \theta^-)$. For a window $(x^{\theta^-}, x^{\theta^+})$ the number G of Definition VI.2 is x^{ξ}. For $w \geq \xi$ let

$$\mathscr{A}_w = (\xi, w] \cup [\theta^+, \infty),$$

and let ϖ_w be the function on $\mathbb{R}$ that vanishes on $(-\infty, \xi]$ and satisfies

$$r\,\varpi_w(r) = \mathbf{1}_{\mathscr{A}_w}(r) + \int_{\mathscr{A}_w \cap (0,r)} \varpi_w(r - z)\,dz \qquad (r > \xi). \quad (25)$$

The right side involves ϖ_w only on $(-\infty, r - \xi)$, so (25) determines ϖ_w on $(\xi, 2\xi]$, then on $(2\xi, 3\xi]$, and so on. Let ρ be Dickman's function, with $\rho(v) = 1$ for $0 \leq v \leq 1$ and $v\rho'(v) = -\rho(v-1)$ for $v > 1$. For $s_1, s_2 > \theta^-$ put $U(s_1, s_2) = 0$ if s_1 or s_2 lies in (θ^-, θ^+), and otherwise

$$U(s_1, s_2) = \iint_{\Delta} \rho(t_1/\xi)\,\rho(t_2/\xi)\,\varpi_w(s_1 - t_1)\,\varpi_w(s_2 - t_2)\,dt_1\,dt_2, \quad (26)$$

where $\Delta = \{t_1, t_2 \geq 0,\ t_1 + t_2 \leq \theta^-\}$ and $w = \max(\theta^- - t_1 - t_2, \xi)$. Then $0 \leq U \leq 1$ (Appendix A 7).

The function U is the density of the pairs that fail the rule. A number u of size x^{s_1} splits as $u = T(u)\,r$, where the x^{ξ}-smooth part $T(u)$ has size x^{t_1} with density $\rho(t_1/\xi)$, and the rough part r has prime factors x^{z_i} with $z_i \geq \xi$, distributed with density $\prod_i dz_i/z_i$ by Mertens' theorem.[8] The pair fails (R1) and (R2) when $s_1, s_2 \notin (\theta^-, \theta^+)$ and no z_i lies in (θ^-, θ^+), and it fails (R3) when $t_1 + t_2 \leq \theta^-$ and every z_i in $[\xi, \theta^-]$ is at most $\theta^- - t_1 - t_2$. These conditions say that all the z_i lie in $\mathscr{A}_w$.

Lemma VI.4. *Let $0 < \theta^- < \theta^+$, let U be the function (26), and let $R \subset (\theta^-, K)^2$, for some K, be a set whose boundary has Lebesgue measure zero. Let $h : \mathbb{N}^2 \to \mathbb{R}$ satisfy $\sum_{d,e} |h(d,e)|/(de) < \infty$, and put*

$$W(u, v) = \sum_{d \mid u,\, e \mid v} h(d, e), \qquad H = \sum_{d,e} \frac{h(d,e)}{de}.$$

Then, as $x \to \infty$,

$$\sum_{(u,v)} \frac{W(u,v)}{uv} = H(\log x)^2 \iint_R (1 - U)\,ds_1\,ds_2 + o\big((\log x)^2\big),$$

where (u, v) runs over the pairs that pass the rule for $(x^{\theta^-}, x^{\theta^+})$ and satisfy $(\log u, \log v)/\log x \in R$.

The proof is in Appendix A 7.

D. The pieces

Fix $\delta \in (0, \frac{1}{200}]$ and a partition

$$\delta = \vartheta_0 < \vartheta_1 < \cdots < \vartheta_m = \frac{1}{42} - 2\delta, \qquad \vartheta_{i+1} - \vartheta_i \leq \delta.$$

For $0 \leq i < m$ put

$$\vartheta_i^+ = \vartheta_{i+1} + \delta, \qquad \theta_i^- = 2\vartheta_i^+ + \delta, \qquad \theta_i^+ = \theta(\vartheta_i^+) - \delta.$$

Since $\vartheta_i^+ \leq \frac{1}{42} - \delta$, we have $\frac{1}{10} - \frac{17}{5}\vartheta_i^+ \geq \frac{2}{105}$ and $\frac{1}{2} - 21\vartheta_i^+ \geq 21\delta$, so $\theta_i^+ - \theta_i^- = \theta(\vartheta_i^+) - 2\vartheta_i^+ - 2\delta > 0$. As before, $x = 2^j$ for the block $I_j = (x, 2x]$.

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[7] If $T(n)$ collected all the prime factors below A, the integer 4 and the window $(\frac{5}{2}, \frac{7}{2})$ would pass (R3) with $T(4) = 4 > \frac{5}{2}$, while none of the divisors $1, 2, 4$ lies in the window. A brute-force check finds this failure 12,022,742 times among the integers up to $2 \cdot 10^6$ and 64 windows, and no failure of Lemma VI.3 (Section XI).

[8] For a number counted with the weight $1/n$ the logarithmic sizes of its prime factors above x^{ξ} behave like the points of a Poisson process with intensity dz/z on $[\xi, \infty)$, and its x^{ξ}-smooth part is independent of them; this is the content of the theorems of Mertens and of Dickman and de Bruijn [23, Chapter III.5]. Equation (25) is the renewal equation for the density of $\sum_i z_i$ when every z_i is restricted to $\mathscr{A}_w$; with $\mathscr{A}_w$ replaced by (ξ, ∞) it is Buchstab's equation.

Definition VI.5. For a prime $p \equiv 3 \pmod 4$ in I_j and $0 \le i < m$, let $\mathscr{Z}^A_{p,i}$ be the set of triples $t = (u,a,\alpha) \in \mathscr{T}(p)$ with

$$x^{1/2+\vartheta_i} < ua \le x^{1/2+\vartheta_{i+1}}, \qquad u,a > x^{\theta_i^-},$$

such that (u,a) passes the rule for the window $(x^{\theta_i^-}, x^{\theta_i^+})$. Let $\mathscr{Z}^B_{p,i}$ be defined in the same way with the pair (u,α) in place of (u,a), and let $\mathscr{Z}^C_{p,i} = \iota(\mathscr{Z}^B_{p,i})$, defined with the pair (a,α).

Lemma VI.6. *Let j be large in terms of δ and $p \in I_j$. Then every piece $\mathscr{Z}^K_{p,i}$ lies in the centre $\mathscr{Z}_p$. In $\mathscr{Z}^A_{p,i}$ the largest coordinate is α, in $\mathscr{Z}^B_{p,i}$ it is a, and in $\mathscr{Z}^C_{p,i}$ it is u. The pieces $\mathscr{Z}^K_{p,i}$, for $K \in \{A,B,C\}$ and $0 \le i < m$, are pairwise disjoint.*

Proof. Let $t \in \mathscr{Z}^A_{p,i}$. Since $x < p \le 2x$ we have $x/4 < P < x$. Since $\theta_i^- = 2\vartheta_{i+1} + 3\delta$,

$$u = \frac{ua}{a} < x^{1/2+\vartheta_{i+1}-\theta_i^-} = x^{1/2-\vartheta_{i+1}-3\delta},$$

so $u^2 < x^{1-8\delta} < x/4 < P$ for j large, and in the same way $a^2 < P$. Next $\alpha = P/(ua) < Px^{-1/2}$, so $\alpha^2 < P \cdot P/x < P$. Hence $t \in \mathscr{Z}_p$. Moreover $\alpha \ge P/x^{1/2+\vartheta_{i+1}} > \frac{1}{4}x^{1/2-\vartheta_{i+1}}$, which exceeds u and a for j large. The pieces $\mathscr{Z}^B_{p,i}$ and $\mathscr{Z}^C_{p,i}$ are treated in the same way. Pieces with different K are disjoint because their largest coordinates differ, and pieces with the same K and different i are disjoint because the ranges of the relevant product are disjoint. □

Proposition VI.7. *For $K \in \{A,B,C\}$ and $0 \le i < m$, as $j \to \infty$,*

$$\sum_{p \in I_j} \frac{|\mathscr{Z}^K_{p,i}|}{p} = \kappa(\log 2)^2 \mathscr{M}_i\, j + o(j),$$

where $\mathscr{M}_i = \iint_{R_i} (1-U_i)\, ds_1\, ds_2$, with

$$R_i = \Big\{ (s_1,s_2) : \tfrac{1}{2} + \vartheta_i < s_1 + s_2 \le \tfrac{1}{2} + \vartheta_{i+1},\ s_1, s_2 > \theta_i^- \Big\}$$

and U_i the function (26) for the window (θ_i^-, θ_i^+).

Proof. For C the statement is the one for B, since ι is a bijection from $\mathscr{Z}^B_{p,i}$ onto $\mathscr{Z}^C_{p,i}$. Write $\eta = \delta/2$.

The moduli are in Maynard's range. Let (u,a) satisfy the conditions of Definition VI.5 and let $y \in [x,2x]$. The modulus $q = 4ua$ satisfies $q \le 4x^{1/2+\vartheta_{i+1}} \le y^{1/2+\vartheta_i^+}$ for j large. By Lemma VI.3, ua has a divisor d with $x^{\theta_i^-} < d < x^{\theta_i^+}$, and for j large

$$y^{2\vartheta_i^+ + \eta} < x^{\theta_i^-}, \qquad x^{\theta_i^+} < y^{\theta(\vartheta_i^+) - \eta}.$$

Since $1 - 42\vartheta_i^+ \ge 42\delta$, we have $\eta < (1-42\vartheta_i^+)/4$. Hence $q \in Q_{\vartheta_i^+,\eta}(y)$.

The piece A. A coprime pair (u,a) as above gives a triple of $\mathscr{Z}^A_{p,i}$ exactly when $p \equiv -1 \pmod{4ua}$, and each modulus $4n$ arises from at most $\tau(n)$ pairs. Theorem VI.1 with these multiplicities, applied at the points $y_l = x(1 + l/j)$ for $0 \le l \le j$, and partial summation over the subintervals $(y_l, y_{l+1}]$ (Appendix A 4) give

$$\sum_{p \in I_j} \frac{|\mathscr{Z}^A_{p,i}|}{p} = \Big(\sum_{p \in I_j} \frac{1}{p} \Big) \sum_{(u,a)} \frac{1}{\varphi(4ua)} + O(1),$$

where (u,a) runs over the coprime pairs that pass the rule with $(\log u, \log a)/\log x \in R_i$. By (A1), $1/\varphi(4ua) = W_A(u,a)/(2ua)$ with $W_A(u,a) = \psi(ua)\mathbf{1}[\gcd(u,a)=1]$, which has the form of Lemma VI.4 with $H = 2\kappa$ (Appendix A 1). Lemma VI.4 gives $\sum_{(u,a)} 1/\varphi(4ua) = \kappa(j\log 2)^2 \mathscr{M}_i + o(j^2)$, and $\sum_{p \in I_j} 1/p = 1/j + O(1/j^2)$.

The piece B. A pair (u,α) gives a triple of $\mathscr{Z}^B_{p,i}$ when $p \equiv -1 \pmod{4u\alpha}$ and $\gcd(u,a) = 1$ with $a = (p+1)/(4u\alpha)$. As in Proposition V.2, inclusion and exclusion over $d \mid (u, P_z)$ with $z = j^{1/2}$ replaces this by the progressions modulo $4u\alpha d$. These moduli are at most $4x^{1/2+\vartheta_{i+1}} e^{2z} \le y^{1/2+\vartheta_i^+}$ and keep the divisor of $u\alpha$ found by Lemma VI.3, so they lie in $Q_{\vartheta_i^+,\eta}(y)$, and a modulus $4n$ arises from at most $\tau_3(n)$ triples (u,α,d). The main term carries the weight $\psi(u\alpha)\prod_{q \mid u, q \le z}(1-1/q)$, which differs from $W_B(u,\alpha) = \psi(u\alpha)\varphi(u)/u$ by a total of $O(j^2/z)$ over the piece (Appendix A 5), and W_B has $H = 2\kappa$. The pairs with a common prime factor $q > z$ of u and a contribute $O(j/z) + o(1)$ (Appendix A 6). The rest is as for A. □

E. The density of all the classes

For $p \in I_j$ with j large in terms of δ, let $\omega_\delta(p)$ be the number of distinct classes $-\mathrm{c}(t)$ with t in one of the three corners at level X_j or in one of the pieces $\mathscr{Z}^K_{p,i}$, and let $\omega_\delta(p) = 0$ for all other primes.

Corollary VI.8. *As $\lambda \to \infty$, $N[\omega_\delta](\lambda) = (a_\delta + o(1))\lambda^2$ with*

$$a_\delta = \frac{3\kappa}{4}\Big(\frac{1}{2} - \delta\Big)^2 + \frac{3\kappa}{2} \sum_{i<m} \mathscr{M}_i.$$

Proof. The corners at level $X_j \le \sqrt{p}/4$ are pairwise disjoint (Lemma II.5), the pieces lie in the centre and are pairwise disjoint (Lemma VI.6), and the centre is disjoint from the corners (Lemma IV.2). The classes are injective on each corner and on the centre, and the only coincidences between different regions are those counted by $\nu(p)$ and $\nu'(p)$ (Propositions III.1, IV.3 and IV.4). Hence

$$\Sigma_p - \nu(p) - \nu'(p) \le \omega_\delta(p) \le \Sigma_p,$$

where Σ_p is the sum of $|K_p(X_j)|$ over the three corners and of $|\mathscr{Z}^K_{p,i}|$ over the pieces. Propositions V.2 and VI.7, summed over the blocks $j \le J$, give

$$\sum_{p \le 2^{J+1}} \frac{\Sigma_p}{p} = a_\delta\, (J\log 2)^2 + o(J^2),$$

and the coincidences contribute $O(1)$ by Corollaries III.3 and IV.5. □

Lemma VI.9. *For $0<\vartheta<\frac{1}{42}$ let U_ϑ be the function (26) for the window $(2\vartheta,\theta(\vartheta))$, and put*

$$\ell(\vartheta)=\int_{2\vartheta}^{1/2-\vartheta}U_\vartheta(s,\tfrac12+\vartheta-s)\,ds,\qquad \mathscr{J}=\int_0^{1/42}\ell(\vartheta)\,d\vartheta. \tag{27}$$

Then $\sum_i \mathscr{M}_i\to\frac{13}{1176}-\mathscr{J}$ as $\delta\to0$, uniformly over the partitions of mesh at most δ. Consequently

$$a_\delta\to a_1=\frac{\kappa}{4}\Big(\frac{40}{49}-6\mathscr{J}\Big).$$

Proof. The union of the regions R_i is the set of (s_1,s_2) with $\vartheta=s_1+s_2-\frac12\in(\delta,\frac{1}{42}-2\delta]$ and $s_1,s_2>\theta^-_{i(\vartheta)}$, where $\theta_i^-=2\vartheta+O(\delta)$. In the coordinates (s_1,ϑ) its area tends to

$$\int_0^{1/42}\Big(\frac12-3\vartheta\Big)d\vartheta=\frac{1}{84}-\frac{3}{3528}=\frac{13}{1176},$$

and $\sum_i\iint_{R_i}U_i$ tends to $\mathscr{J}$ by dominated convergence, because the windows (θ_i^-,θ_i^+) converge to $(2\vartheta,\theta(\vartheta))$ (Appendix A 7). Finally $\frac{3\kappa}{16}+\frac{3\kappa}{2}(\frac{13}{1176}-\mathscr{J})=\frac{\kappa}{4}(\frac{40}{49}-6\mathscr{J})$. □

VII. THE LARGE SIEVE AND THE TILT

Let $\omega(p)$ be integers with $0\le\omega(p)<p$, and for each prime p let Ω_p be a set of $\omega(p)$ residue classes modulo p. Put $g(p)=\omega(p)/(p-\omega(p))$ and

$$L(Q)=\sum_{q\le Q}\mu^2(q)\prod_{p\mid q}g(p). \tag{28}$$

Lemma VII.1 (Large sieve [17, 18])**.** *For all $N,Q\ge1$, the number of $n\le N$ with $n \bmod p\notin\Omega_p$ for every prime $p\le Q$ is at most $(N+Q^2)/L(Q)$.*

The size of $L(Q)$ is determined by the density of the classes. We assume $\omega(p)\le C_\eta p^\eta$ for every $\eta>0$; this holds in our applications because $\omega(p)\le|\mathscr{T}(p)|\le\tau_3(P)$. Let $N(\lambda)=N[\omega](\lambda)$ as in (23).

Proposition VII.2. *Let $a>0$ and $T=\log Q$.*

(i) For every $\sigma>0$,

$$L(Q)\le Q^\sigma\exp F(\sigma),\qquad F(\sigma)=\sum_p g(p)p^{-\sigma}. \tag{29}$$

If $N(\lambda)\le(a+o(1))\lambda^2$, then $F(\sigma)\le(2a+o(1))\sigma^{-2}$ as $\sigma\to0^+$, and

$$\log L(Q)\le\frac32(4a)^{1/3}T^{2/3}(1+o(1)).$$

(ii) If $N(\lambda)=(a+o(1))\lambda^2$, then

$$\log L(Q)\ge\frac32(4a)^{1/3}T^{2/3}(1-o(1)).$$

Proof. (i) Since $(Q/q)^\sigma\ge1$ for $q\le Q$,

$$L(Q)\le Q^\sigma\sum_q\mu^2(q)\frac{\prod_{p\mid q}g(p)}{q^\sigma}=Q^\sigma\prod_p\big(1+g(p)p^{-\sigma}\big),$$

and $1+y\le e^y$ gives (29); this is Rankin's device [21]. The difference $g(p)-\omega(p)/p=\omega(p)^2/(p(p-\omega(p)))$ is summable, and partial summation gives $\sum_p(\omega(p)/p)p^{-\sigma}=\sigma\int_0^\infty e^{-\sigma\lambda}N(\lambda)\,d\lambda\le(2a+o(1))\sigma^{-2}$ (Appendix A 9). The choice $\sigma^3T=4a$ minimises $\sigma T+2a/\sigma^2$, with value $\frac32\sigma T=\frac32(4a)^{1/3}T^{2/3}$.

(ii) Fix $\sigma>0$ and let X_p, for $p\le Q$, be independent random variables with $X_p=1$ with probability $\pi_p=y_p/(1+y_p)$, where $y_p=g(p)p^{-\sigma}$, and $X_p=0$ otherwise. The random squarefree number $q=\prod_p p^{X_p}$ takes the value $q\mid\prod_{p\le Q}p$ with probability $\prod_{p\mid q}y_p/\prod_{p\le Q}(1+y_p)$. Let μ_σ and V_σ be the mean and variance of $\log q=\sum_p X_p\log p$. Suppose $\mu_\sigma+\sqrt{2V_\sigma}=T$. By Chebyshev's inequality, $|\log q-\mu_\sigma|<\sqrt{2V_\sigma}$ with probability at least $\frac12$, and then $q\le Q$ and $q^\sigma\ge e^{\sigma(T-2\sqrt{2V_\sigma})}$. Hence

$$\begin{aligned}L(Q)&\ge\sum_{\substack{q\le Q\\ |\log q-\mu_\sigma|<\sqrt{2V_\sigma}}}\mu^2(q)\frac{\prod_{p\mid q}g(p)}{q^\sigma}q^\sigma\\&\ge\frac12 e^{\sigma T-2\sigma\sqrt{2V_\sigma}}\prod_{p\le Q}(1+y_p).\end{aligned} \tag{30}$$

In Appendix A 10 we show that such a σ exists, that $\sigma T\to\infty$ and $\sigma\asymp T^{-1/3}$, that $V_\sigma\ll\sigma^{-4}$, and that $\sum_{p\le Q}\log(1+y_p)\ge(2a-o(1))\sigma^{-2}$. Then (30) gives

$$\log L(Q)\ge\sigma T+\frac{2a-o(1)}{\sigma^2}-O(T^{1/3}),$$

and $\sigma T+2a'/\sigma^2\ge\frac32(4a')^{1/3}T^{2/3}$ for every $\sigma>0$ and $a'>0$. □

The lower bound (30) is an exponential tilt: the weight $q^{-\sigma}$ moves the distribution of $\log q$ so that its mean sits at $\log Q$, and the sum of the large sieve is then comparable to the whole Euler product.[9]

VIII. PROOFS OF THE THEOREMS

Proof of Theorem A. Fix $\delta\in(0,\frac{1}{200}]$ and a partition of mesh at most δ as in Section VI. For $p\in I_j$ with j large, take Ω_p to be the set of the $\omega_\delta(p)$ classes $-\mathrm{c}(t)$ with t in the corners at level X_j or in the pieces $\mathscr{Z}^K_{p,i}$, and take Ω_p empty otherwise.

[9] The same picture explains the shape of the bound. The Euler product grows like $\exp(2a/\sigma^2)$, the tilt costs $Q^{-\sigma}$, and the balance $\sigma\asymp(\log Q)^{-1/3}$ gives the exponent $(\log Q)^{2/3}$. With $Q=\sqrt{N}$ the factor $(\frac12)^{2/3}$ combines with $(4a)^{1/3}$ to give $a^{1/3}$, which is why the final constant is $\frac32a^{1/3}$.

By Lemma II.2, every $n \in \mathscr{E}(N)$ avoids these classes for all p. Lemma VII.1 with $Q = \sqrt{N}$ gives

$$|\mathscr{E}(N)| \le \frac{2N}{L_\delta(\sqrt{N})}.$$

By Corollary VI.8 and Proposition VII.2(ii) with $T = \frac{1}{2}\log N$,

$$\log L_\delta(\sqrt{N}) \ge \frac{3}{2}(4a_\delta)^{1/3}\Big(\frac{\log N}{2}\Big)^{2/3}(1-o(1)).$$

Since $(4a_\delta)^{1/3}2^{-2/3} = a_\delta^{1/3}$, this is $\frac{3}{2}a_\delta^{1/3}(\log N)^{2/3}(1-o(1))$. By Lemma VI.9, $\frac{3}{2}a_\delta^{1/3} \to \frac{3}{2}a_1^{1/3} = c_1$ as $\delta \to 0$, since $\kappa/4 = 9/(8\pi^2)$. Given $\varepsilon > 0$, choose δ with $\frac{3}{2}a_\delta^{1/3} > c_1 - \varepsilon/2$; then (3) holds for all large N. The bound (5) is Proposition IX.1, and $\mathscr{J} \ge 0$ because $U \ge 0$. □

Theorem B needs only an upper bound for the density of the classes $\Omega_p^\sharp$, which come from triples with one product at most $\sqrt{p}$. Let $\omega^\sharp(p) = |\Omega_p^\sharp|$ and $N^\sharp(\lambda) = N[\omega^\sharp](\lambda)$.

Proposition VIII.1. $N^\sharp(\lambda) \le (a_0 + o(1))\lambda^2$.

Proof. We have $\omega^\sharp(p) \le |A_p^\sharp| + |B_p(\sqrt{p})| + |C_p(\sqrt{p})|$, where $A_p^\sharp$ is $A_p(\sqrt{p})$ together with the triples with $u = 1$ or $a = 1$ and $ua \le \sqrt{p}$. Fix $\delta > 0$ and split each corner according to whether the relevant product is at most $p^{1/2-\delta}$ or lies in $(p^{1/2-\delta}, p^{1/2}]$. The first parts are counted as in Proposition V.2, with the upper bound $|B_p| \le |B_p^{(z)}|$ and the level $(2^{j+1})^{1/2-\delta}$ on the block I_j; they contribute $(a_\delta^{\mathrm{cor}} + o(1))\lambda^2$. For the second parts we drop the condition $\gcd(u,a) = 1$ and use the Brun–Titchmarsh inequality, which loses a constant factor but applies to a range of measure $O(\delta)$; they contribute $O(\delta\lambda^2)$ (Appendix A 11). Since δ is arbitrary and $a_\delta^{\mathrm{cor}} \le a_0$, the proposition follows. □

Proof of Theorem B. Let $T_0 = \frac{1}{2}\log N$ and $\sigma = (4a_0/T_0)^{1/3}$. By Propositions VIII.1 and VII.2(i), for N large,

$$\sigma T_0 + F^\sharp(\sigma) \le \frac{3}{2}(4a_0)^{1/3}T_0^{2/3}(1+\varepsilon') = (c_0+\varepsilon)(\log N)^{2/3},$$

with ε' chosen in terms of ε. If $Q \le \sqrt{N}$, then $L^\sharp(Q) \le L^\sharp(\sqrt{N}) \le e^{\sigma T_0 + F^\sharp(\sigma)}$, and $(N+Q^2)/L^\sharp(Q) \ge N/L^\sharp(\sqrt{N})$. If $Q > \sqrt{N}$, then by (29) and $\sigma < 2$,

$$\frac{N+Q^2}{L^\sharp(Q)} \ge Q^{2-\sigma}e^{-F^\sharp(\sigma)} \ge N^{1-\sigma/2}e^{-F^\sharp(\sigma)} = Ne^{-\sigma T_0 - F^\sharp(\sigma)}.$$

In both cases the bound of the theorem follows. □

IX. THE VALUE OF THE CONSTANT

Proposition IX.1. $\mathscr{J} \le \frac{533}{64680}$. *Hence* $\frac{40}{49} - 6\mathscr{J} \ge \frac{1181}{1540}$ *and* $c_1 \ge c_{\mathrm{var}}$, *with* $c_{\mathrm{var}}^3 = 286983/(98560\pi^2)$.

Proof. By the definition (26), $U_\vartheta(s_1,s_2) = 0$ when s_1 or s_2 lies in the window $(2\vartheta, \theta(\vartheta))$, and $U_\vartheta \le 1$ always. On the segment $s_1 + s_2 = \frac{1}{2} + \vartheta$ with $s_1, s_2 > 2\vartheta$, both lie outside the window exactly when $\theta(\vartheta) \le s_1 \le \frac{1}{2} + \vartheta - \theta(\vartheta)$. Hence

$$\ell(\vartheta) \le \Big(\frac{1}{2} + \vartheta - 2\theta(\vartheta)\Big)_+.$$

On $(0, \frac{1}{44}]$ the right side is $\frac{3}{10} + \frac{19}{5}\vartheta$, and on $(\frac{1}{44}, \frac{1}{42})$ it is $39\vartheta - \frac{1}{2}$; both are positive. Integrating,

$$\int_0^{1/44}\Big(\frac{3}{10} + \frac{19\vartheta}{5}\Big)d\vartheta = \frac{151}{19360},$$
$$\int_{1/44}^{1/42}\Big(39\vartheta - \frac{1}{2}\Big)d\vartheta = \frac{251}{569184},$$

with sum $\frac{533}{64680}$. Then $\frac{40}{49} - \frac{6\cdot 533}{64680} = \frac{1181}{1540}$, and $c_1^3 \ge \frac{27}{8}\cdot\frac{9}{8\pi^2}\cdot\frac{1181}{1540} = \frac{286983}{98560\pi^2}$. □

The bound of Proposition IX.1 uses only (R1). To include (R2) and (R3) we compute $\mathscr{J}$ numerically. For each depth ϑ we solve (25) on a grid of step $\xi/16$, tabulate Dickman's function and Buchstab's function on a grid of step $2\cdot 10^{-5}$, and evaluate (26) and the two outer integrals of (27) by Gauss–Legendre quadrature, with the nodes split at the points where the integrand has kinks, among them $\vartheta = \frac{1}{54}$ and $\vartheta = \frac{1}{44}$. This gives

$$\mathscr{J} = 0.000306, \qquad c_1 = 0.67921.$$

A second program, written independently, solves (25) on the same grid and evaluates the integrals by the midpoint rule; it gives $\mathscr{J} = 0.000300$ and $c_1 = 0.67922$, and its loss ℓ agrees with the first to within 0.75% at five depths. A Monte Carlo simulation, which draws the prime factors of u and v by stick breaking and applies the rule of Definition VI.2 directly, agrees with the computed values of U at four points to within 0.002. The value $c_1 = 0.6792$ is therefore stable to four decimals, while the proved bound is $c_1 \ge c_{\mathrm{var}} = 0.66571$.

The rules (R1) alone, (R1) with (R2), and (R1) with (R2) and the condition that the part of uv made of primes up to A lies in (A,B), give the losses shown in Figure 3(b) and the constants 0.66571, 0.67760 and 0.67831. Almost all the remaining loss of the full rule comes from depths ϑ near $\frac{1}{42}$, where the window of Maynard's theorem closes.

X. REMARKS

Remark X.1 (Smaller and larger regions)**.** The argument applies to any union of corners, and the density is proportional to the area covered (Table I and Figure 4). A single corner gives $c = \frac{3}{2}(\kappa/16)^{1/3} = 0.4582$, two corners give $\frac{3}{2}(\kappa/8)^{1/3} = 0.5773$, and the three corners give c_0. Each row with strips includes the three corners; the share of the strips counted by (R1) alone is the closed form $\frac{1181}{1540}$ of Proposition IX.1, and the share of all of the strips is $\frac{40}{49}$.

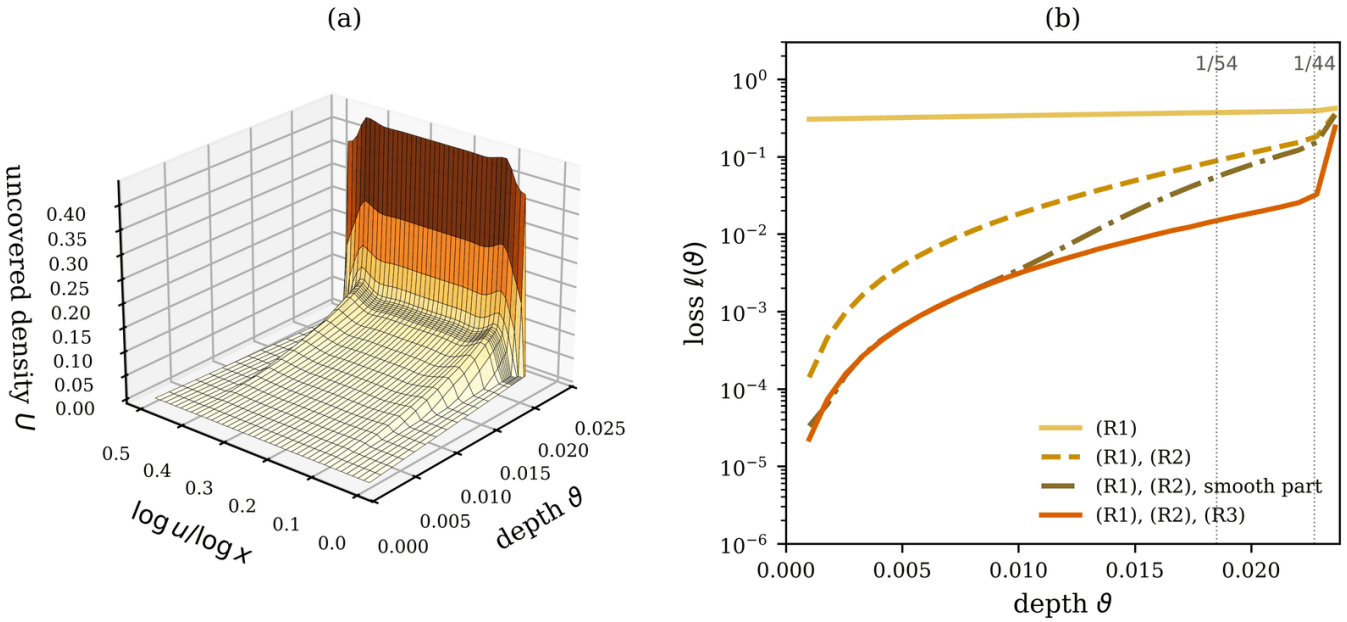


FIG. 3. (a) The density U of the pairs that fail the rule of Definition VI.2, on the segments $s_1+s_2=\frac{1}{2}+\vartheta$, as a function of $s_1=\log u/\log x$ and of the depth ϑ. It vanishes where u or v lies in Maynard's window and is largest near $\vartheta=\frac{1}{42}$. (b) The loss $\ell(\vartheta)$ of (27) for the rule (R1) alone, for (R1) with (R2), for an intermediate rule that also accepts the part of uv made of primes up to A when it lies in the window, and for the full rule (R1)–(R3). The dotted lines mark $\vartheta=\frac{1}{54}$ and $\vartheta=\frac{1}{44}$, where the window changes its shape.

TABLE I. The constant given by the large sieve for regions of the triangle of Figure 1. The density is $a=$ (share of the triangle)$\cdot\kappa/4$, and $c=\frac{3}{2}a^{1/3}$. The rows with strips count them through the rules of Definition VI.2; the row with all of the strips would need a divisor in Maynard's range for every modulus, and the last row would need moduli beyond that range.

Region	Share	a	c
One corner	0.25	0.0284966	0.4582
Two corners	0.5	0.0569932	0.5773
Three corners	0.75	0.0854897	0.6608
Strips by (R1)	0.766883	0.0874142	0.6657
Strips by (R1), (R2)	0.808696	0.0921802	0.6776
Strips by (R1)–(R3)	0.814491	0.0928408	0.6792
All of the strips	0.816327	0.0930501	0.6797
Whole triangle	1	0.1139863	0.7273

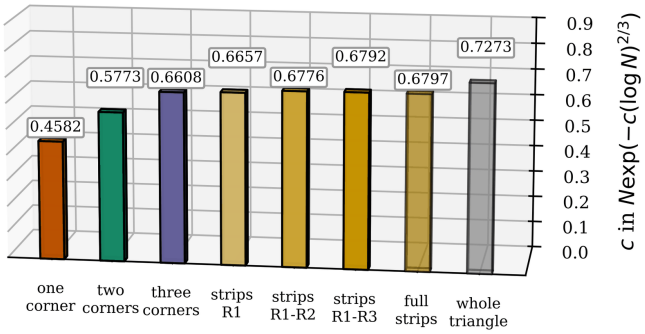


FIG. 4. The constant c in $N\exp(-c(\log N)^{2/3})$ for one, two and three corners, for the corners with the strips counted by the three rules, for the corners with all of the strips, and for the whole triangle, which is not available.

Remark X.2 (The level)**.** Section III uses the level $16X^2\le p$. The factor 16 does not affect the constants, since only $\log X/\log p\to\frac{1}{2}$ enters the density, but the bounds (15) do depend on it. For the primes $p\equiv 3 \pmod 4$ below 10^6 no two triples in the corners at level $\sqrt{p}/4$ share a class, while at level $\sqrt{p}/2$ there are 17,568 such pairs, all between B and C. The first is at $p=31$, where $(1,4,2)\in B_{31}(2)$ and $(8,1,1)\in C_{31}(2)$ both have the class 8; here $P=8$, far below the bound $P\ge 1024$ of (15).

Remark X.3 (Where the method stops)**.** The strips have depth $\frac{1}{42}$ because the range of divisors in Theorem VI.1 is empty for $\vartheta\ge\frac{1}{42}$. If every triple of the strips were counted, the share would be $\frac{3}{4}+6(\frac{1}{84}-\frac{3}{3528})=\frac{40}{49}$ and the constant $\frac{3}{2}(45/(49\pi^2))^{1/3}=0.6797$; the rule of Definition VI.2 loses $6\mathscr{J}=0.0018$ of this share. A larger constant needs primes in a fixed residue class to moduli beyond $p^{1/2+1/42}$. Under the Elliott–Halberstam conjecture [9] the whole centre could be counted through the smallest of the three products, which is at most $P^{2/3}$; since the classes of the centre are distinct and differ from those of the corners apart from the pairs counted by $\nu'(p)$ (Propositions IV.3 and IV.4), the whole triangle would then give $\frac{3}{2}(\kappa/4)^{1/3}=0.7273$.

Remark X.4 (Vaughan's count of classes)**.** In the form given in [20, Section 4], Vaughan's argument uses at each

prime $f(p)=\lfloor\frac12\sum_{t\mid P}|\mu(t)|\tau(P/t)\rfloor$ classes. The sum equals $\sum_{d\mid P}2^{\omega(d)}$, the number of Type II triples, so $f(p)$ is half of $|\mathscr{T}(p)|$. If the primes are equidistributed in progressions to moduli up to $p^{2/3}$, the number of triples has density $\kappa/4$, the whole triangle, and the classes $f(p)$ have density at most $\kappa/8$. By Proposition VII.2(i) and the argument of Theorem B, no sieve with $f(p)$ classes at each prime can then give a constant above $\frac32(\kappa/8)^{1/3}=0.5773$, below both c_0 and c_1.

XI. FORMAL AND NUMERICAL VERIFICATION

The algebraic steps are checked in Lean 4 [12] with Mathlib [14], in 48 theorems. For the corners: the normal forms (8); the identities (9) and (10); the identities (17), (11), (12) and (14) as consequences of their hypotheses; the three inequalities (15); the relation $\varphi(qn)=q\,\varphi(n)$ for $q\mid n$; the local factors of S_B; the optimisation in Proposition VII.2; and the value $c_0^3=729/(256\pi^2)$. For the centre: the Farey bound and the bound $4u^2\alpha<p+1$ of Proposition IV.3; the identities (20)–(22), the inequality $4ks\le a$ and the case $j=0$ of Proposition IV.4; and the family of Remark IV.6, with the primality of 531359 and 1442783. For the strips: the range of Theorem VI.1 and the values $\frac{1}{44}$ and $\frac{1}{54}$; the divisor chain of Lemma VI.3; the area $\frac{13}{1176}$ and the share $\frac{40}{49}$. All these theorems depend at most on the axioms `propext`, `Classical.choice` and `Quot.sound`. The analytic estimates of Sections V, VI and VII are not formalised.

Independent programs in C, C++, Python and Julia check the remaining steps numerically. Over the 332,398 primes $p\equiv3 \pmod 4$ below 10^7, with 37,093,810 Type II triples, the corners at level $\sqrt{p}/16$ are disjoint, the three normal forms agree, the classes in each corner are distinct, and there is no coincidence at all between different corners. The same primes have 6,943,856 triples in the centre, of which 1,760,868 have their largest coordinate at least $P^{1/2-1/42}$. The classes of the centre are distinct; they agree with no class of A at level $\sqrt{p}/4$, with no class of B for $u<a$ and no class of C for $u>a$; and the remaining coincidences are the four pairs of Remark IV.6. The divisor chain of Lemma VI.3 holds in all 128,000,000 cases of the integers up to $2\cdot10^6$ with 64 windows. The sums $\sum_p|A_p|/p$ and $\sum_p|B_p|/p$ agree with the predictions $\sum_p S_A/p$ and $\sum_p S_B/p$ to within 0.25%. The parametrisation of Proposition III.2 is checked by brute force against all coincidences with $P\le4000$, without the restriction to the corners. The mean values of Lemma V.1 are computed to $X=2\cdot10^6$, where $(S-\frac{\kappa}{2}\log^2X)/\log X$ settles near -0.47 for S_A and 0.55 for S_B. The two bounds of Proposition VII.2 are evaluated for the density a_0 up to $T=10^{15}$, where their ratio to the exact optimum is 1.0000. The integral $\mathscr{J}$ is computed as in Section IX.

The engine runs. In the first run the library of Anaxia held 22 typed construction nodes: the large sieve with the tilted lower bound; each corner paired with each of the three normal forms, or left out; the centre, left out or counted under the Elliott–Halberstam conjecture; and the inputs that each of these needs. Typed crossing produced 128 closed constructions, which the engine tested on the 39,322 primes $p\equiv3 \pmod 4$ below 10^6. It kept 7, those with one, two or three corners and the matching normal forms; it rejected 112 because a normal form failed to separate the classes, 8 that contain the centre because a class of the centre equals a class of a corner at $p=531359$ (Remark IV.6), and the empty construction. The best of the seven gives c_0. In the second run the library held 19 nodes for the centre: the three normal forms; five rules for finding a divisor of the modulus in Maynard's range, from (R1) alone to the rule of Definition VI.2, and a rule that omits the bound on the steps of the chain; the family of coincidences of Proposition IV.4, allowed or not; and the Elliott–Halberstam conjecture. Of its 37 closed constructions it accepted 5, with the constants 0.660789 (centre left out), 0.665710, 0.677596, 0.678311 and 0.679210, marked one as conditional (the whole centre under the Elliott–Halberstam conjecture, 0.727292), and rejected 31: 24 because the forms $4u^2\alpha$ and $(4a^2\alpha)^{-1}$ do not separate the classes of the centre (they fail on all 12,399 primes tested), 6 because they do not allow the two coincidences of the family below 10^6, and the rule without the bound on the steps, which claims a divisor that does not exist 1,449,392 times among the integers up to $2\cdot10^5$ and 64 windows, first at $n=4$. The best construction is the one of this paper. The records of both runs and the scripts that reproduce them are in the archive [25].

Appendix A: Detailed derivations

1. The local factors

The totient of $4n$. If n is odd then $\varphi(4n)=\varphi(4)\varphi(n)=2\varphi(n)$. If $n=2^\beta n'$ with $\beta\ge1$ and n' odd then $\varphi(4n)=2^{\beta+1}\varphi(n')=4\varphi(n)$. In both cases

$$\frac{1}{\varphi(4n)}=\frac{\psi(n)}{2n},\qquad \psi(n)=\prod_{q\mid n,\ q\text{ odd}}\frac{q}{q-1}. \tag{A1}$$

For a prime q dividing n, $\varphi(qn)=q\,\varphi(n)$.

The corner B. With (A1),

$$S_B(X)=\frac12\sum_{u\alpha\le X}\frac{W_B(u,\alpha)}{u\alpha},\qquad W_B(u,\alpha)=\frac{\varphi(u)}{u}\,\psi(u\alpha).$$

The function W_B is multiplicative in the pair: $W_B(u,\alpha)=\prod_q W_B(q^i,q^j)$ with $q^i\parallel u$ and $q^j\parallel\alpha$. At an odd prime q its local values are $W_B=1$ at $(0,0)$, $W_B=(1-\frac1q)\frac{q}{q-1}=1$ for $i\ge1$, and $W_B=\frac{q}{q-1}$ for $i=0$, $j\ge1$. At $q=2$ the factor ψ is absent and $W_B=\frac12$ for $i\ge1$, $W_B=1$ for $i=0$.

The local generating function at an odd prime q is

$$\sum_{i,j\ge0}W_B(q^i,q^j)x^iy^j=1+\frac{x}{(1-x)(1-y)}+\frac{q}{q-1}\cdot\frac{y}{1-y},$$

and multiplying by $(1-x)(1-y)$ gives

$$G_q(x,y)=1+\frac{y}{q-1}-\frac{xy}{q-1}.$$

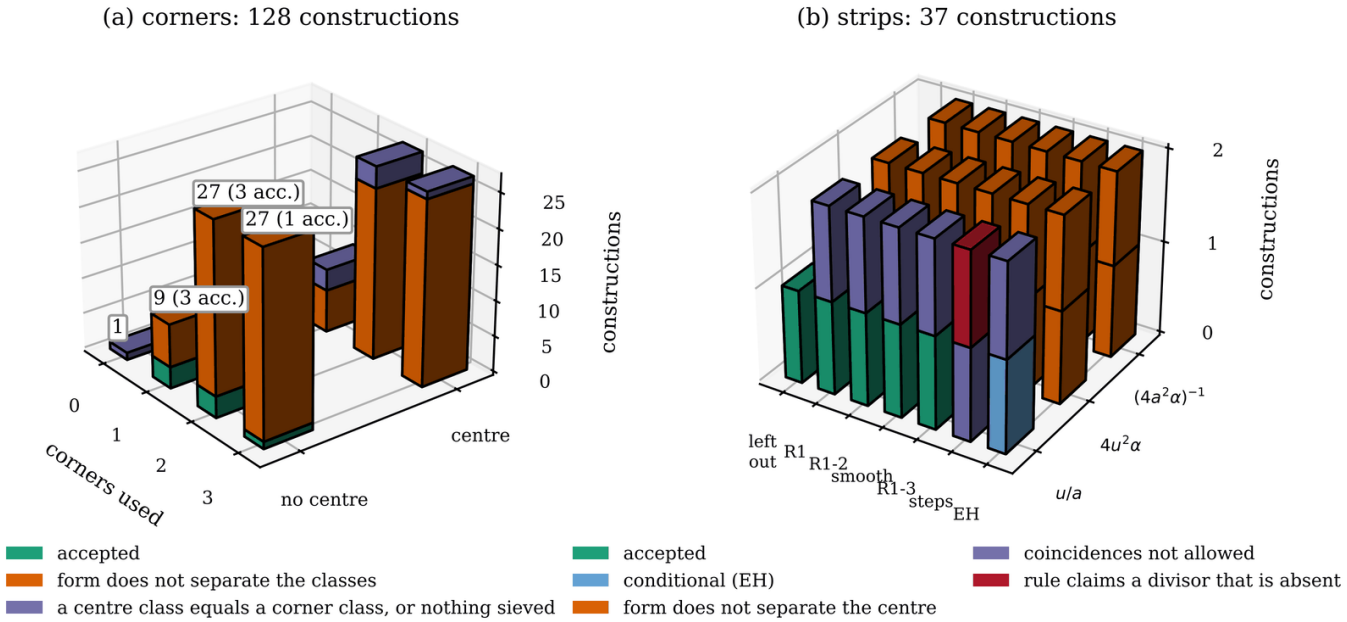


FIG. 5. The two engine runs. (a) The 128 constructions of the corners, by the number of corners and by whether the centre is included. Seven are accepted; the rest are rejected because a normal form does not separate the classes, because a class of the centre equals a class of a corner, or because they sieve nothing. (b) The 37 constructions of the strips, by the rule that finds the divisor and by the normal form used in the centre, with the family of coincidences allowed (front) or not (back). Five are accepted, and the whole centre under the Elliott–Halberstam conjecture is conditional.

At $q=2$ the same computation gives $G_2(x,y)=1-\frac{x}{2}$. Hence $W_B(u,\alpha)=\sum_{d|u,e|\alpha} g(d,e)$, where g is multiplicative in the pair with the local values read off from G_q: $g(1,q)=\frac{1}{q-1}$ and $g(q,q)=-\frac{1}{q-1}$ for odd q, $g(2,1)=-\frac{1}{2}$, and $g=0$ at all other prime powers except $g(1,1)=1$. At $x=y=1/q$,

$$G_q\Big(\frac{1}{q},\frac{1}{q}\Big)=1+\frac{1}{q(q-1)}-\frac{1}{q^2(q-1)}=1+\frac{1}{q^2},$$

and $G_2(\frac{1}{2},\frac{1}{2})=\frac{3}{4}$. The same value $1+q^{-2}$ is the quotient of the local sum $\sum_{i,j} W_B(q^i,q^j)q^{-i-j}=(q^2+1)/(q-1)^2$ by $(1-1/q)^{-2}$.

The corner A. Here

$$S_A(X)+R(X)=\frac{1}{2}\sum_{ua\le X}\frac{W_A(u,a)}{ua},$$

where $W_A(u,a)=\psi(ua)$ when $\gcd(u,a)=1$, $W_A=0$ otherwise, and $R(X)$ is the sum over the pairs with $u=1$ or $a=1$. Since $\sum_{n\le X}1/\varphi(4n)=O(\log X)$, $R(X)=O(\log X)$. The local values of W_A at an odd prime are 1 at $(0,0)$, $\frac{q}{q-1}$ when exactly one exponent is positive, and 0 when both are; at $q=2$ they are $1,1,0$. The local generating functions after multiplication by $(1-x)(1-y)$ are

$$G_q^A(x,y)=1+\frac{x+y}{q-1}-\frac{(q+1)xy}{q-1},\qquad G_2^A=1-xy,$$

with $G_q^A(\frac{1}{q},\frac{1}{q})=1+q^{-2}$ and $G_2^A(\frac{1}{2},\frac{1}{2})=\frac{3}{4}$. The coefficients give $W_A=\sum_{d|u,e|a}h_A(d,e)$ with $\sum_{d,e}|h_A(d,e)|/(de)<\infty$, as Lemma VI.4 requires, and the same holds for g.

TABLE II. The local factors of the two mean values. Both corners give $1+q^{-2}$ at odd q and $\frac{3}{4}$ at 2.

	$G(x,y)$ at odd q	at $x=y=\frac{1}{q}$	$q=2$
B	$1+\frac{y-xy}{q-1}$	$1+q^{-2}$	$\frac{3}{4}$
A	$1+\frac{x+y-(q+1)xy}{q-1}$	$1+q^{-2}$	$\frac{3}{4}$

The constant. The Euler product of the values at $x=y=1/q$ is

$$\frac{3}{4}\prod_{q\ \text{odd}}\Big(1+\frac{1}{q^2}\Big)=\frac{3}{4}\cdot\frac{4}{5}\cdot\frac{\zeta(2)}{\zeta(4)}=\frac{3}{4}\cdot\frac{4}{5}\cdot\frac{15}{\pi^2}=\frac{9}{\pi^2}=2\kappa,$$

using $\zeta(2)=\pi^2/6$ and $\zeta(4)=\pi^4/90$.

2. The hyperbola sum

For $Y\ge 1$ let $H(Y)=\sum_{mn\le Y}1/(mn)$ and $H(Y)=0$ for $Y<1$. With $h(x)=\sum_{m\le x}1/m=\log x+\gamma+O(1/x)$,

$$H(Y)=\sum_{m\le Y}\frac{1}{m}h\Big(\frac{Y}{m}\Big)=\sum_{m\le Y}\frac{\log Y-\log m+\gamma}{m}+O\Big(\frac{\log Y}{Y}\Big).$$

Since $\sum_{m\le Y}(\log m)/m=\frac{1}{2}\log^2 Y+O(1)$, this gives $H(Y)=\frac{1}{2}\log^2 Y+2\gamma\log Y+O(1)$.

Write $\Lambda=\log X$. Then, with g as in Appendix A 1,

$$2S_B(X)=\sum_{de\le X}\frac{g(d,e)}{de}\Big(\frac12(\Lambda-\log de)^2+O(1+\Lambda)\Big).$$

The series $\sum_{d,e}|g(d,e)|(1+\log de)^2/(de)$ converges, since its Euler factor at an odd q is $1+O(q^{-2}\log^2 q)$. For the pairs with $de>X$, $\sum_{de>X}|g(d,e)|/(de)\le X^{-1/4}\sum_{d,e}|g(d,e)|(de)^{-3/4}$, and the last series converges because its terms at q are $O(q^{-7/4})$ and $O(q^{-5/2})$. Expanding the square,

$$2S_B(X)=\frac{\Lambda^2}{2}\sum_{d,e}\frac{g(d,e)}{de}+O(\Lambda)=\kappa\Lambda^2+O(\Lambda).$$

For S_A the coefficients of G_q^A are of size q^{-1} at $(1,0)$ and $(0,1)$ and of size 1 at $(1,1)$, so the terms of $\sum|h_A(d,e)|(de)^{-3/4}$ at q are $O(q^{-7/4})$ and $O(q^{-3/2})$; the rest is identical.

3. The Bombieri–Vinogradov theorem with multiplicities

The Bombieri–Vinogradov theorem [5], [7, Chapter 28] states that for every $A>0$ there is $B>0$ such that

$$\sum_{q\le x^{1/2}(\log x)^{-B}}\max_{(b,q)=1}\max_{y\le x}\Big|\pi(y;q,b)-\frac{\pi(y)}{\varphi(q)}\Big|\ll_A\frac{x}{(\log x)^A}.$$

Let $E(q)$ be the inner maximum. In the proof of Proposition V.2 each modulus $q=4n$ occurs with multiplicity at most $\tau_3(n)$. Using the trivial bound $E(q)\ll x/q$, Cauchy's inequality and $\sum_{n\le x}\tau_3(n)^2/n\ll(\log x)^9$,

$$\sum_q\tau_3(q)E(q)\le\Big(\sum_q\tau_3(q)^2E(q)\Big)^{1/2}\Big(\sum_qE(q)\Big)^{1/2}\ll\frac{x}{(\log x)^{(A-9)/2}}.$$

Taking $A=15$ gives the error $O(x(\log x)^{-3})$ used in the proof. The inclusion–exclusion over $d\mid(u,P_z)$ replaces each pair by the progressions modulo $4u\alpha d$ with $d\mid u$; a modulus $4m$ arises from at most $\tau_3(m)$ triples (u,α,d), so the same estimate covers all of them. The moduli $4u\alpha d\le 4X_je^{2z}$, since $\prod_{q\le z}q\le e^{2z}$, are below $x^{1/2}(\log x)^{-B}$ for j large because $X_j=2^{j(1/2-\delta)}$ and $z=j^{1/2}$.

4. Maynard's theorem with multiplicities

In the proof of Proposition VI.7, fix $y\in[x,2x]$ and let $E(q;y)=|\pi(y;q,-1)-\pi(y)/\varphi(q)|$. For $q\le y$ we have $E(q;y)\le\pi(y;q,-1)+\pi(y)/\varphi(q)\ll y\log\log y/q$, and each modulus q that occurs lies in $Q_{\vartheta_i^+,\eta}(y)$ and occurs at most $\tau_3(q/4)$ times. Since $\sum_{n\le y}\tau_3(n)^2/n\ll(\log y)^9$, the sum of $\tau_3(q/4)^2E(q;y)$ over these moduli is $O(y(\log y)^{10})$, and by Theorem VI.1 with $A=17$ the sum of $E(q;y)$ is $O(y(\log y)^{-17})$. Cauchy's inequality gives

$$\sum_q\tau_3(q/4)\,E(q;y)\ll\frac{y}{(\log y)^{7/2}}.$$

Let $f(p)=|\mathscr{Z}_{p,i}^{\mathscr{P}A}|$ and $F(y)=\sum_{x<p\le y}f(p)$, and let S be the sum of $1/\varphi(4ua)$ over the pairs (u,a) of the proof of Proposition VI.7. Applying the bound at y and at x gives $F(y)=(\pi(y)-\pi(x))S+O(x(\log x)^{-7/2})$ at each of the $j+1$ points $y_l=x(1+l/j)$. On $(y_l,y_{l+1}]$ the weight $1/p$ lies between $1/y_{l+1}$ and $1/y_l$, whose ratio is at most $1+1/j$, and $f\ge0$, so

$$\sum_{y_l<p\le y_{l+1}}\frac{f(p)}{p}=\frac{F(y_{l+1})-F(y_l)}{y_l}\Big(1+O\Big(\frac1j\Big)\Big).$$

The same holds with f replaced by S times the indicator of the primes. Summing over l,

$$\sum_{p\in I_j}\frac{f(p)}{p}=S\sum_{p\in I_j}\frac1p\Big(1+O\Big(\frac1j\Big)\Big)+O\big(j(\log x)^{-7/2}\big).$$

Since $S\ll j^2$ and $\sum_{p\in I_j}1/p\ll1/j$, the error is $O(1)$. The piece B is treated in the same way with the moduli $4u\alpha d$.

5. Large prime factors of $\gcd(u,a)$

The truncated product. For $u\ge1$,

$$0\le\prod_{\substack{q\mid u\\ q\le z}}\Big(1-\frac1q\Big)-\frac{\varphi(u)}{u}\le\sum_{\substack{q\mid u\\ q>z}}\frac1q.$$

Writing $u=qu'$ and using $\varphi(4qu'\alpha)\ge(q-1)\varphi(4u'\alpha)$,

$$S_B^{(z)}(X)-S_B(X)\le\sum_{q>z}\frac{1}{q(q-1)}\sum_{u'\alpha\le X}\frac{1}{\varphi(4u'\alpha)}\ll\frac{\log^2X}{z},$$

because $\sum_{n\le X}\tau(n)/\varphi(4n)\ll\log^2X$. The same bound, with $X=x^K$, controls the difference of the weights in the piece B of Proposition VI.7.

The pairs with a common prime $q>z$. Such a pair satisfies $p\equiv-1\pmod{4u\alpha q}$ with $q\mid u$, and $m=4u\alpha q\le 4X_j^2\le4\cdot2^{j(1-2\delta)}$. For $x=2^{j+1}$ we have $\log(x/m)\ge\delta j\log2$ when $j\ge3/\delta$. The Brun–Titchmarsh inequality $\pi(x;m,b)\le 2x/(\varphi(m)\log(x/m))$ [18] gives

$$\pi(x;m,-1)\le\frac{2\cdot2^{j+1}}{\varphi(m)\,\delta j\log2}\le\frac{5\,\pi(I_j)}{\delta\,\varphi(m)}$$

for j large, since $\pi(I_j)\sim2^j/(j\log2)$. Summing over $q>z$ and the pairs, with $\varphi(4u\alpha q)=q\,\varphi(4u\alpha)$, the total over $p\in I_j$ is at most

$$\frac{5\,\pi(I_j)}{\delta}\sum_{q>z}\frac{1}{q(q-1)}\sum_{u'\alpha\le X_j}\frac{1}{\varphi(4u'\alpha)}\ll\frac{\pi(I_j)\log^2X_j}{\delta z}.$$

With the weight $1/p\le2^{-j}$ this contributes $O(j/(\delta z))$ to $\sum_{p\in I_j}|B_p|/p$.

6. Large common prime factors in the strips

In the piece B of Proposition VI.7 the pairs (u,α) satisfy $u\alpha \le x^{1/2+1/42}$. A pair whose u and a share a prime $q > z$ satisfies $p \equiv -1 \pmod m$ with $m = 4u\alpha q$, $q \mid u$. If $q \le x^{1/3}$, then $m \le 4x^{1/2+1/42+1/3}$, so $\log(2x/m) \ge \frac{1}{8}\log x$, and the Brun–Titchmarsh inequality gives $\pi(2x;m,-1) \ll x/(\varphi(m)\log x)$. As in Appendix A.5, the sum over $q > z$ and the pairs is $\ll (x/\log x)(\log x)^2/z$, and with the weight $1/p \le 1/x$ it contributes $O(j/z)$. If $q > x^{1/3}$, we use $\pi(2x;m,-1) \le 2x/m+1$. Writing $u = qu'$, the first term gives

$$\sum_{q>x^{1/3}} \frac{2x}{4q^2} \sum_{u'\alpha\le x} \frac{1}{u'\alpha} \ll x^{2/3}(\log x)^2,$$

and the second term is at most the number of triples (q,u',α) with $qu'\alpha \le x^{1/2+1/42}$, which is $O(x^{0.53})$. With the weight $1/p \le 1/x$ both are $o(1)$.

7. The density of the pairs that fail

Profiles. Fix $\xi > 0$. For $x > 1$ and $n \ge 1$, the ξ-profile of n is the pair $\Pi_x(n) = (t,Z)$, where $t = \log T(n)/\log x$, with $T(n)$ the product of the prime powers of n whose prime is below x^ξ, and Z is the multiset of the numbers $\log q/\log x$ over the prime factors $q \ge x^\xi$ of n, counted with multiplicity. The size of n is $s = t + \sum Z$. On the space of profiles let

$$\lambda_1 = \rho(t/\xi)\,dt \otimes \nu, \qquad \nu = \delta_\varnothing + \sum_{k\ge1} \frac{1}{k!} \prod_{i=1}^{k} \frac{\mathbf{1}[z_i \ge \xi]\,dz_i}{z_i},$$

where ν is a measure on finite multisets $\{z_1,\dots,z_k\}$. Every n factors uniquely as $n = T(n)\,r$ with r free of primes below x^ξ, and $1/n = (1/T(n))(1/r)$. By the theorem of Dickman and de Bruijn [23, Chapter III.5], $\Psi(X,X^{1/v}) = X\rho(v)(1+o(1))$ uniformly for bounded v, and partial summation gives

$$\sum_{\substack{T\le x^t \\ T\ x^\xi\text{-smooth}}} \frac{1}{T} = \log x \int_0^t \rho\Big(\frac{\tau}{\xi}\Big)\,d\tau + o(\log x)$$

uniformly for bounded t. By Mertens' theorem,

$$\sum_{x^z<q\le x^{z'}} \frac{1}{q} = \log\frac{z'}{z} + o(1) \qquad (\xi \le z < z'),$$

and the numbers r with a repeated prime factor $q \ge x^\xi$ have total weight $O(x^{-\xi})$ relative to the others. Hence, for every bounded set F of profiles whose boundary is λ_1-null,

$$\frac{1}{\log x} \sum_{n:\ \Pi_x(n)\in F} \frac{1}{n} \to \lambda_1(F).$$

Since the weight $1/(uv)$ is a product, the same holds for pairs: $(\log x)^{-2}\sum_{(\Pi_x(u),\Pi_x(v))\in F} 1/(uv) \to \lambda_1\otimes\lambda_1(F)$ for every bounded F with $\lambda_1\otimes\lambda_1$-null boundary. The image of λ_1 under the size s is Lebesgue measure, since $\sum_{n\le x^s} 1/n = s\log x + O(1)$.

The failure set. Let $\xi = \min(\theta^+ - \theta^-, \theta^-)$, so that $G = x^\xi$ for the window $(x^{\theta^-}, x^{\theta^+})$. The pair (u,v) fails the rule exactly when its profiles (t_1,Z_1), (t_2,Z_2) satisfy: $s_1,s_2 \notin (\theta^-,\theta^+)$; no element of $Z_1\cup Z_2$ lies in (θ^-,θ^+); and the largest element of $(Z_1\cup Z_2)\cap[\xi,\theta^-]$, or 0 if there is none, plus t_1+t_2 is at most θ^-. When $s_1,s_2 > \theta^-$, the last two conditions say that $t_1 + t_2 \le \theta^-$ and that every element of $Z_1\cup Z_2$ lies in $\mathscr{A}_w$ with $w = \max(\theta^- - t_1 - t_2, \xi)$, up to the λ_1-null event that an element equals ξ. The boundary of the failure set lies in the union of the sets where s_i, an element of $Z_1\cup Z_2$, or the sum in the last condition takes one of the values ξ, θ^-, θ^+, which are $\lambda_1\otimes\lambda_1$-null on $\{s_1,s_2 > \theta^-\}$. For fixed t_i, the density of $\sum Z_i$ over the multisets in $\mathscr{A}_w$ is ϖ_w: writing f for this density, $rf(r)$ is the ν-integral of $\sum_i z_i$ over the multisets with sum r, and by the symmetry of ν this equals $\mathbf{1}_{\mathscr{A}_w}(r) + \int_{\mathscr{A}_w\cap(0,r)} f(r-z)\,dz$, which is (25). The empty multiset would give $s_i = t_i \le \theta^-$, which is excluded. Hence the $\lambda_1\otimes\lambda_1$-measure of the failing profiles with $(s_1,s_2)\in R$ is $\iint_R U$.

The bound $0 \le U \le 1$. Let ϖ be the solution of (25) with $\mathscr{A}_w$ replaced by (ξ,∞), the density of $\sum Z$ over all multisets. Then $0 \le \varpi_w \le \varpi$, and dropping the condition $t_1+t_2 \le \theta^-$ in (26),

$$U(s_1,s_2) \le \prod_{i=1}^{2} \int_0^{s_i} \rho\Big(\frac{t}{\xi}\Big)\varpi(s_i - t)\,dt \le 1,$$

because each factor is the λ_1-density at s_i of the profiles with a nonempty multiset, and the total density is 1.

Proof of Lemma VI.4. For $h = \mathbf{1}_{(1,1)}$ the statement is the convergence above, applied to the passing profiles with $(s_1,s_2)\in R$, which form a bounded set with null boundary and have measure $\iint_R(1-U)$. For general h,

$$\sum_{(u,v)} \frac{W(u,v)}{uv} = \sum_{d,e} \frac{h(d,e)}{de} \sum_{(u',v')} \frac{1}{u'v'},$$

where (u',v') runs over the pairs for which (du',ev') passes and has its sizes in R. For fixed d,e and x^ξ larger than every prime factor of de, the profiles of du' and ev' are those of u' and v' with t increased by $\log d/\log x$ and $\log e/\log x$, which tend to 0; since the limit set has null boundary, the inner sum is $(\log x)^2\iint_R(1-U) + o((\log x)^2)$. It is also at most $\sum_{u',v'\le x^K} 1/(u'v') \ll (\log x)^2$, so dominated convergence over (d,e) with $\sum|h(d,e)|/(de) < \infty$ gives the lemma.

Proof of the limit in Lemma VI.9. Fix $\eta_0 > 0$. For $\vartheta \in [\eta_0, \frac{1}{42} - \eta_0]$ the number $\xi(\vartheta) = \min(\theta(\vartheta) - 2\vartheta, 2\vartheta)$ is at least $\xi_0 = \min(\frac{2}{105}, 2\eta_0)$, and the same holds for the windows (θ_i^-,θ_i^+) with $\vartheta_i \in [\eta_0, \frac{1}{42}-\eta_0]$ once δ is small. The ξ-profile of a number is a function of its ξ_0-profile, so all these failure sets live in the space of ξ_0-profiles with the measure $\lambda_1\otimes\lambda_1$ at level ξ_0, and $\sum_i \iint_{R_i} U_i$, restricted to these depths, is the measure of the set of profile pairs with $\vartheta = s_1 + s_2 - \frac{1}{2}$ in the range and (s_1,s_2) in $\bigcup_i R_i$ that fail the rule for the window of their piece. As $\delta \to 0$ these windows converge to $(2\vartheta,\theta(\vartheta))$, and outside a null set of profiles the indicator of failure converges to that for the limiting window. The sets lie in a fixed

bounded region of finite measure, so by dominated convergence the measure tends to $\int_{\eta_0}^{1/42-\eta_0}\ell(\vartheta)\,d\vartheta$. The depths outside $[\eta_0,\frac{1}{42}-\eta_0]$ contribute $O(\eta_0)$ to both sides, because $0\le U\le 1$, and η_0 is arbitrary.

8. Partial summation in a block

Let $f(p)\ge 0$ and $F(y)=\sum_{2^j<p\le y}f(p)$ for $y\in I_j$. Suppose $F(y)=(\pi(y)-\pi(2^j))S+E(y)$ with $|E(y)|\le E$ for all $y\in I_j$. Then

$$\sum_{p\in I_j}\frac{f(p)}{p}=\frac{F(2^{j+1})}{2^{j+1}}+\int_{2^j}^{2^{j+1}}\frac{F(y)}{y^2}\,dy$$
$$=S\sum_{p\in I_j}\frac{1}{p}+O\Big(\frac{E}{2^j}\Big),$$

because the same identity holds for $\pi(y)-\pi(2^j)$ in place of F and $\int_{2^j}^{2^{j+1}}y^{-2}\,dy=2^{-j-1}$. With $E\ll 2^j(\log 2^j)^{-3}$ the error is $O(j^{-3})$. The pairs removed in Appendix A 5 are handled with the trivial weight $1/p\le 2^{-j}$. Finally, the prime number theorem with the error term $O(y\exp(-c\sqrt{\log y}))$ gives $\sum_{p\le y}1/p=\log\log y+M_1+O(\exp(-c\sqrt{\log y}))$, so $\sum_{p\in I_j}1/p=\log(1+1/j)+O(e^{-c\sqrt{j}})=1/j+O(1/j^2)$.

9. From the density to the Euler product

Let $N(\lambda)=\sum_{p\le e^\lambda}\omega(p)/p$ and $G_0(\sigma)=\sum_p(\omega(p)/p)p^{-\sigma}$. Since $N(\lambda)=O(\lambda^2)$, integration by parts gives

$$G_0(\sigma)=\int_0^\infty e^{-\sigma\lambda}\,dN(\lambda)=\sigma\int_0^\infty e^{-\sigma\lambda}N(\lambda)\,d\lambda.$$

If $N(\lambda)\le(a+\varepsilon)\lambda^2+C_\varepsilon$, then $G_0(\sigma)\le(a+\varepsilon)\sigma\cdot 2\sigma^{-3}+C_\varepsilon=2(a+\varepsilon)\sigma^{-2}+C_\varepsilon$. In the same way

$$\sum_p\frac{\omega(p)\log p}{p^{1+\sigma}}=\int_0^\infty N(\lambda)(\sigma\lambda-1)e^{-\sigma\lambda}\,d\lambda,$$
$$\sum_p\frac{\omega(p)\log^2 p}{p^{1+\sigma}}=\int_0^\infty N(\lambda)(\sigma\lambda^2-2\lambda)e^{-\sigma\lambda}\,d\lambda,$$

and, when $N(\lambda)=(a+o(1))\lambda^2$, these are $(4a+o(1))\sigma^{-3}$ and $(12a+o(1))\sigma^{-4}$ as $\sigma\to 0^+$, from $\int_0^\infty\lambda^2(\sigma\lambda-1)e^{-\sigma\lambda}d\lambda=4\sigma^{-3}$ and $\int_0^\infty\lambda^2(\sigma\lambda^2-2\lambda)e^{-\sigma\lambda}d\lambda=12\sigma^{-4}$. If the sums are restricted to $p\le Q=e^T$, the same asymptotics hold provided $\sigma T\to\infty$, since the tails are bounded by $\sigma^{-k}\int_{\sigma T}^\infty x^{k+1}e^{-x}dx$ with $k=2,3,4$. Finally $0\le g(p)-\omega(p)/p\le 2\omega(p)^2/p^2$ once $\omega(p)\le p/2$, which holds for all large p because $\omega(p)\le C_\eta p^\eta$; hence $F(\sigma)=G_0(\sigma)+O(1)$, and the same holds for the sums with $\log p$ and $\log^2 p$.

10. The choice of σ

Assume $N(\lambda)=(a+o(1))\lambda^2$ with $a>0$. With $y_p=g(p)p^{-\sigma}$ and $\pi_p=y_p/(1+y_p)$,

$$\mu_\sigma=\sum_{p\le Q}\pi_p\log p,\qquad V_\sigma=\sum_{p\le Q}\pi_p(1-\pi_p)\log^2 p.$$

Both are continuous in σ, and both tend to 0 as $\sigma\to\infty$ because the sums are finite. Since $y_p-y_p^2\le\pi_p\le y_p$, Appendix A 9 gives, whenever $\sigma T\to\infty$,

$$\mu_\sigma=(4a+o(1))\sigma^{-3},\qquad V_\sigma\le(12a+o(1))\sigma^{-4}.$$

Take $\sigma_1=\varepsilon_0T^{-1/3}$ with $\varepsilon_0>0$ small. Then $\sigma_1T\to\infty$ and $\mu_{\sigma_1}\ge 3a\varepsilon_0^{-3}T>T$, so by continuity there is $\sigma\ge\sigma_1$ with $\mu_\sigma+\sqrt{2V_\sigma}=T$. If σT stayed bounded along a sequence of T, say $\sigma T\le K$, then $p^{-\sigma}\ge e^{-K}$ for $p\le Q$ and

$$\mu_\sigma\ge\frac{e^{-K}}{2}\sum_{p\le Q}\frac{\omega(p)\log p}{p}\gg e^{-K}T^3,$$

contradicting $\mu_\sigma\le T$ for large T. Hence $\sigma T\to\infty$, the asymptotics apply, and $\mu_\sigma\le T$ gives $\sigma\ge(4a-o(1))^{1/3}T^{-1/3}$, while $\sigma\ge\sigma_1$ already gives $\sigma\gg T^{-1/3}$ from below and $T=\mu_\sigma+\sqrt{2V_\sigma}\ll\sigma^{-3}$ gives $\sigma\ll T^{-1/3}$. Therefore $\sigma\sqrt{2V_\sigma}\ll\sigma^{-1}\ll T^{1/3}$. Finally

$$\sum_{p\le Q}\log(1+y_p)\ge\sum_{p\le Q}y_p-\frac12\sum_p y_p^2\ge F_Q(\sigma)-O(1),$$

where F_Q is F restricted to $p\le Q$, and $F_Q(\sigma)=(2a+o(1))\sigma^{-2}$ by Appendix A 9. This is what the proof of Proposition VII.2(ii) uses.

11. The band below $\sqrt{p}$

Proposition VIII.1 uses the following count. For $p\in I_j$ take the pairs (u,α) with $p^{1/2-\delta}<u\alpha\le p^{1/2}$ and $p\equiv -1\pmod{4u\alpha}$; the corners A and C are treated in the same way. Every such pair has $2^{j(1/2-\delta)}<u\alpha\le 2^{(j+1)/2}$, and the modulus $m=4u\alpha$ satisfies $\log(2^{j+1}/m)\ge(\frac{j}{2}-2)\log 2$. The Brun–Titchmarsh inequality gives $\pi(2^{j+1};m,-1)\le(9+o(1))\pi(I_j)/\varphi(m)$, so the number of these pairs, summed over $p\in I_j$, is at most

$$(9+o(1))\,\pi(I_j)\sum_{2^{j(1/2-\delta)}<n\le 2^{(j+1)/2}}\frac{\tau(n)}{\varphi(4n)}.$$

The sum over n is $O(\log^2(2^{j/2})-\log^2(2^{j(1/2-\delta)}))+O(j)=O(\delta j^2)+O(j)$. With the weight $1/p\le 2^{-j}$, the contribution of the block is $O(\delta j)+O(1)$, and the sum over $j\le J$ is $O(\delta J^2)+O(J)=O(\delta\lambda^2)+O(\lambda)$.

DATA AVAILABILITY

The LaTeX source of this paper, the Lean 4 formalisation and the C, C++, Python, Julia and shell programs described in Section XI, the records of the two engine runs and the scripts that reproduce them are available at `https://github.com/creelie/ErdosProblem1` and in the archive [25]. The corners, the treatment of the centre and the rule of Definition VI.2 were selected with the engine Anaxia of the BLLM project [1, 2], version 3.0.5, in the two runs described in Section XI. Deep Bhattacharjee carried out the computations and the Lean, C, C++, Python, Julia and shell verifications, and the proofs and the text were drafted, with AI assistance (Claude, Anthropic); both authors approved the final version.

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